

CHAPTER 5

APPLICATION AND VERIFICATION

INTRODUCTION

In Chapter 2 the origins and key features of the two general activated sludge models included in the simulation package (UCTOLD and IAWPRC) were described briefly. In this Chapter the predictive power of the two models will be evaluated. To test the predictive power of the two models, data sets for a number of activated sludge systems were selected from the experimental database generated over the past 20 years at UCT; the responses of the selected systems were simulated using both models, and the predictions compared to the experimental data. In the simulations, the values for the kinetic and stoichiometric constants were the default values listed in Chapter 2, Tables 2.2 and 2.5, unless explicitly stated otherwise. The screens in the computer programs of the default values are as follows:

For UCTOLD, heterotroph kinetic (at 20° C) and Arrhenius temperature constants,

**** HETEROTROPHS ****		
Mue max	d-1	3.200
Ks COD	(Ksh) g COD m-3	5.000
Ks O2	(Koh) g O2 m-3	0.002
B decay	(bh) d-1	0.620
Neta (growth)		0.330
Ks NO3	(Kno) g N m-3	0.100
Hydrolysis rate	(Kmp) d-1	1.350
Ks hydrolysis	(Ksp) g COD g-1 COD	0.027
Ammonification	(Kr) m3 g-1 COD d-1	0.032
Ks NH3	(Kna) g N m-3	0.010
Adsorption rate	(Ka) g-1 COD m3 d-1	0.170
RETURN TO MENU		

Hit <Arrows> or <SpaceBar> to move selection
Hit <Return> to enter new value for selected constant

** ARRHENIUS TEMP CONSTANTS (Thetas ref 20C) **	
Mue max hetero	1.200
Ksh	1.000
B endogenous hetero (Bh)	1.029
Kmp hydrolysis	1.080
Ksp hydrol. half-sat.	0.910
Kr	1.029
Mue max auto	1.123
Ksa	1.123
B endogenous auto (Ba)	1.029
Ka adsorption rate	1.029
RETURN TO MENU	

Hit <Arrows> or <SpaceBar> to move selection
Hit <Return> to enter new value for selected constant

For IAWPRC, heterotroph kinetic (at 20°C) and Arrhenius temperature constants,

**** HETEROTROPHS ****			
Mue max	d-1		3.200
Ks COD	(Ksh) g COD m-3		5.000
Ks O2	(Koh) g O2 m-3		0.002
B decay	(bh) d-1		0.620
Neta (solubilization)			0.330
Neta (growth)			1.000
Ks NO3	(Kno) g N m-3		0.100
Hydrolysis rate	(Kh) d-1		2.030
Ks hydrolysis	(Kx) g COD g-1 COD		0.027
Ammonification	(Kr) m3 g-1 COD d-1		0.032
Ks NH3	(Kna) g N m-3		0.010
RETURN TO MENU			

Hit <Arrows> or <SpaceBar> to move selection
Hit <Return> to enter new value for selected constant

** ARRHENIUS TEMP CONSTANTS (Thetas ref 20C) **			
Mue max hetero			1.200
Ksh			1.000
B endogenous hetero (Bh)			1.029
Kh hydrolysis			1.080
Kx hydroly. half-sat.			0.910
Kr			1.029
Mue max auto			1.123
Knh			1.123
B endogenous auto (Ba)			1.029
RETURN TO MENU			

Hit <Arrows> or <SpaceBar> to move selection
Hit <Return> to enter new value for selected constant

For UCTOLD and IAWPRC, the autotroph kinetic (at 20°C) and the stoichiometric constants are the same,

**** AUTOTROPHS ****			
Mue max auto	d-1		0.450
Ks NH4+	(Ksa) g N m-3		1.000
Ks O2	(Koa) g O2 m-3		0.002
B endogenous	(ba) d-1		0.040
RETURN TO MENU			

Hit <Arrows> or <SpaceBar> to move selection
Hit <Return> to enter new value for selected constant

**** STOICHIOMETRIC PARAMETERS ****				
Yield, hetero	(Yzh)	g COD g ⁻¹ COD	0.666	
Frac inert	(Fe)	g COD g ⁻¹ COD	0.080	
N in biomass	(Fzb,n)	g N g ⁻¹ COD	0.068	
N in inert	(Fze,n)	g N g ⁻¹ COD	0.068	
Yield, auto	(Yza)	g COD g ⁻¹ COD	0.150	
COD:VSS ratio	(Fcv)	g COD g ⁻¹ VSS	1.480	
Max adsorption	(Fma)	g COD g ⁻¹ COD	1.000	
				RETURN TO MENU

Hit <Arrows> or <SpaceBar> to move selection

Hit <Return> to enter new value for selected constant

The constants characterizing the influent wastewater were determined separately for each system (for procedures see Chapter 6) and are listed where appropriate. The following systems were simulated:

- Aerobic completely mixed reactor systems with constant or cyclic flow and load.
- Aerobic batch tests, with and without nitrification.
- Anoxic/aerobic systems with plugflow primary and secondary anoxic reactors and constant flow and load.
- Anoxic/aerobic systems with completely mixed in-series reactors and constant or cyclic flow and load.
- Contact-stabilization systems with constant or cyclic flow and load.

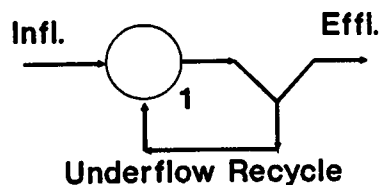
The results obtained for selected typical examples of these simulations are described below.

AEROBIC COMPLETELY MIXED REACTOR SYSTEMS

Simulations were undertaken of aerobic systems with constant and cyclic flow and load influents.

Constant flow and load

Single completely mixed aerobic reactor



The system received unsettled municipal wastewater from Cape Flats, Cape Town, South Africa as influent (Arkley and Marais, 1981). Wastewater characteristics and configuration and operational data as input to the programs were as follows:

**** WASTEWATER CHARACTERISTICS ****		
Sti(avg)	g COD m-3	502.000
Nti(avg)	g N m-3	50.100
Fbs	g COD g-1 COD	0.240
Fs,us	g COD g-1 COD	0.080
Fs,up	g COD g-1 COD	0.130
Fn,a	g N g-1 N	0.750
Fnob,p	g N g-1 N	0.500
Fn,ous	g N g-1 N	0.000
Fs,zbh	g Zbh COD g-1 COD	0.000
VSS/TSS	g VSS g-1 TSS	0.750
Inf Alk	mole m-3	10.000
RETURN TO MENU		

Fbs = readily biodegradable/biodegradable COD
 Fs,us = frac infl COD that is sol unbiodegrad
 Fs,up = frac infl COD that is part unbiodegrad
 Fn,a = frac infl TKN that is NH3/NH4
 Fnob,p = frac organic bio N that is part
 Fn,ous = frac infl TKN that is organic unbio sol
 Fs,zbh = frac infl COD that is heterotrophs

Hit <Arrows>/<SpaceBar>
to move selection

Hit <Return> to enter
new value for
selected constant

**** PROCESS CONFIGURATION ****			
		Gp 1. Number of Reactors = 1	
Gp 2. Reactor Vols, l		Gp 3. Feed Fraction	Gp 4. Aeration/DO
No. 1:	10.00	1.00	3.0

Do you wish to change any parameters?

Y/N....

***** PLANT OPERATING PARAMETERS *****			
1	SRT (total)	d	= 20.0
2	Process Temperature	degC	= 20.0
Flow rates:			
3	Influent flow	l d-1	= 10.0
4	RAS recycle flow	l d-1	= 40.0

Do you wish to change any parameters?

Y/N....

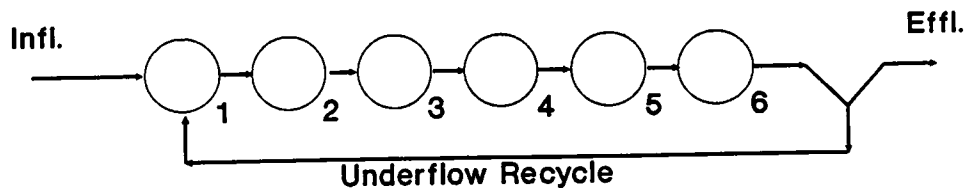
All kinetic and stoichiometric constants were the default values.

The responses predicted by the two models, UCTOLD and IAWPRC, are compared in Table 5.1 with the measured response; evidently good agreement is obtained with both models.

Table 5.1: Comparison of measured and predicted results for completely mixed aerobic single reactor system receiving unsettled wastewater as influent.

Parameter	Reactor			Effluent		
	Measured	UCTOLD	IAWPRC	Measured	UCTOLD	IAWPRC
Nitrate (mgN/ℓ)	39,1	37,6	37,6	39,0	37,6	37,6
VSS (mgVSS/ℓ)	2050	2174	2168	—	—	—
OUR (mgO/ℓ/h)	19,2	20,0	20,0	—	—	—
TKN (mgN/ℓ)	—	—	—	1,5	1,7	1,7
COD (mgCOD/ℓ)	—	—	—	39,3	40,3	41,5

6 in-series completely mixed aerobic reactors



The system received unsettled municipal wastewater from Cape Flats, Cape Town, South Africa as influent (Ekama and Marais, 1979). Data input to the programs was as follows:

**** WASTEWATER CHARACTERISTICS ****		
Sti(avg)	g COD m-3	429.000
Nti(avg)	g N m-3	37.000
Fbs	g COD g-1 COD	0.190
Fs,us	g COD g-1 COD	0.080
Fs,up	g COD g-1 COD	0.130
Fn,a	g N g-1 N	0.750
Fnob,p	g N g-1 N	0.500
Fn,ous	g N g-1 N	0.030
Fs,zbh	g Zbh COD g-1 COD	0.000
VSS/TSS	g VSS g-1 TSS	0.750
Inf Alk	mole m-3	10.000
RETURN TO MENU		

Fbs = readily biodegradable/biodegradable COD Fs,us = frac infl COD that is sol unbiodegrad Fs,up = frac infl COD that is part unbiodegrad Fn,a = frac infl TKN that is NH3/NH4 Fnob,p = frac organic bio N that is part Fn,ous = frac infl TKN that is organic unbio sol Fs,zbh = frac infl COD that is heterotrophs	Hit <Arrows>/<SpaceBar> to move selection Hit <Return> to enter new value for selected constant
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**** PROCESS CONFIGURATION ****      Gp 1. Number of Reactors = 6
      Gp 2. Reactor Vols, l      Gp 3. Feed Fraction      Gp 4. Aeration/DO
No. 1:          2.00                      1.00          3.0
No. 2:          2.00                      3.0
No. 3:          2.00                      3.0
No. 4:          2.00                      3.0
No. 5:          2.00                      3.0
No. 6:          2.00                      3.0

Gp 5. Recycles:  Use to include/remove mixed liquor recycles.
RAS recycle to Reactor No.1
  
```

Do you wish to change any parameters?
Y/N....

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***** PLANT OPERATING PARAMETERS *****
1  SRT (total)      d      =      10.0
2  Process Temperature  degC  =      20.0

Flow rates:
3  Influent flow    l d-1  =      36.0
4  RAS recycle flow l d-1  =      72.0
  
```

Do you wish to change any parameters?
Y/N....

The value for the autotroph maximum specific growth rate ($\hat{\mu}_A$) had to be increased, from the default value of 0,45/d to 0,65/d. Also, since in the system pH = 6,05, the autotroph kinetic constants had to be adjusted (WRC, 1984), as follows:

$$\begin{aligned}
 \hat{\mu}_A &= \hat{\mu}_A (\text{pH} = 7,2) \cdot (2,35)^{(\text{pH}-7,2)} \\
 &= 0,65 \cdot (2,35)^{(6,05-7,2)} \\
 &= 0,243 \text{ /d}
 \end{aligned}
 \tag{5.1}$$

$$\begin{aligned}
 K_{SA} &= K_{SA} (\text{pH} = 7,2) \cdot (2,35)^{(7,2-\text{pH})} \\
 &= 1,0 \cdot (2,35)^{(7,2-6,05)} \\
 &= 2,67 \text{ mgN/l}
 \end{aligned}
 \tag{5.2}$$

All other stoichiometric and kinetic constants used in the simulations were the default values. The predicted and measured results are compared in Fig 5.1(a,b,c,d and e); good agreement is obtained with both models.

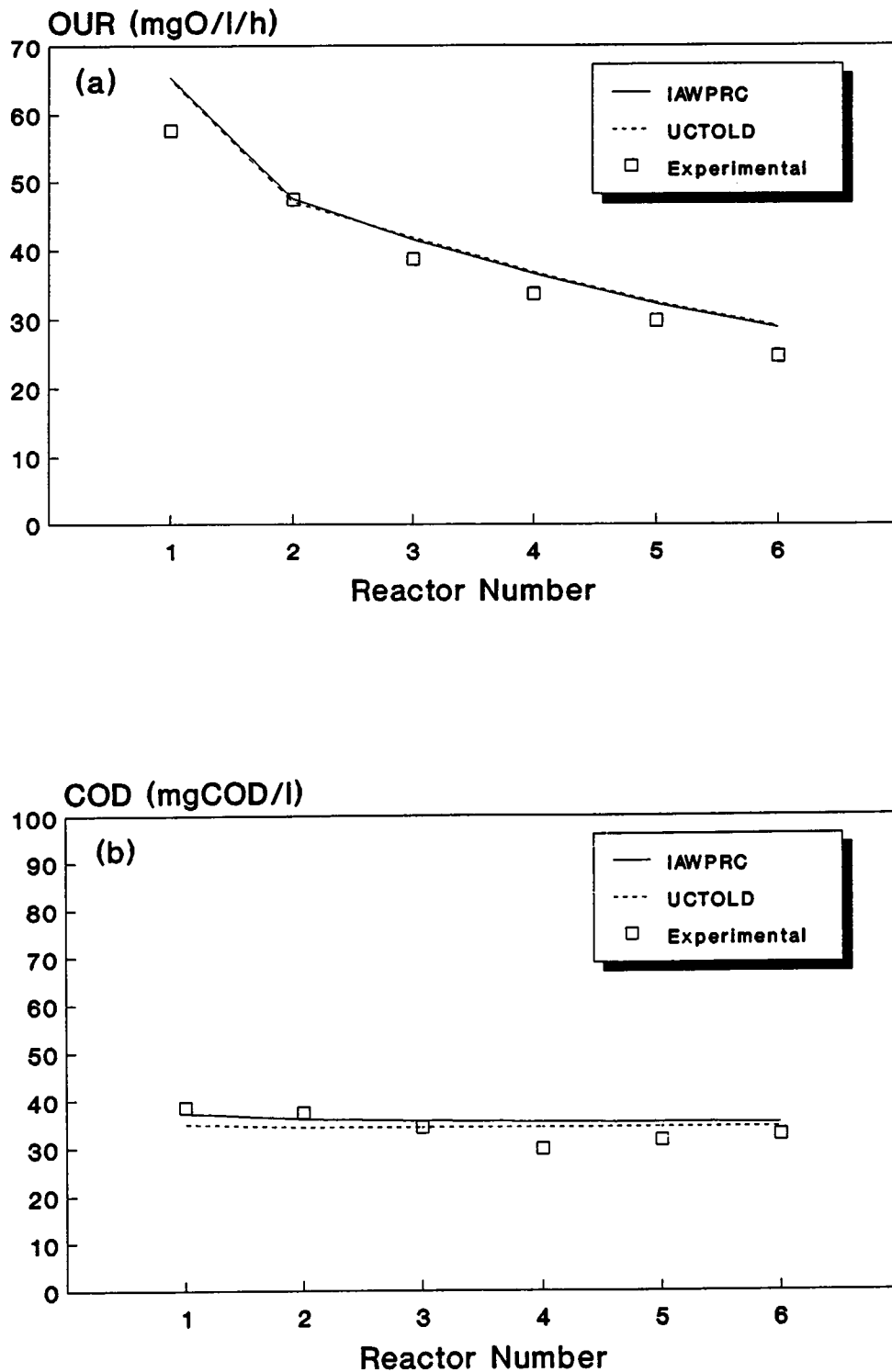


Fig 5.1: Predicted and experimental (a) OUR, (b) COD, (c) VSS, (d) TKN, and (e) nitrate profiles for 6 in-series reactor completely aerobic system under constant flow and load; predictions using UCTOLD and IAWPRC, experimental data from Ekama and Marais (1979). Continued overleaf....

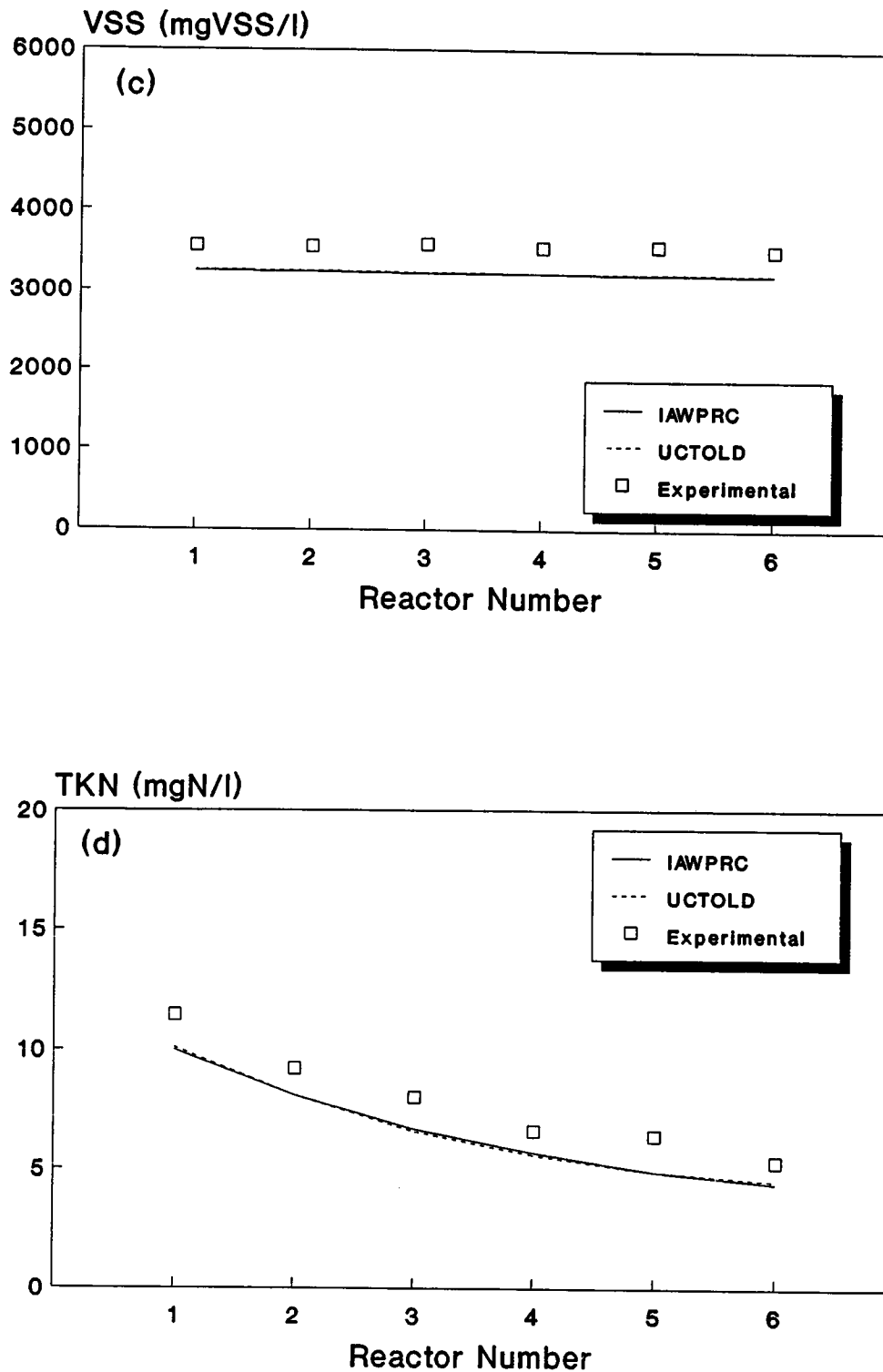


Fig 5.1: Predicted and experimental (a) OUR, (b) COD, (c) VSS, (d) TKN, and (e) nitrate profiles for 6 in-series reactor completely aerobic system under constant flow and load; predictions using UCTOLD and IAWPRC, experimental data from Ekama and Marais (1979). Continued overleaf.....

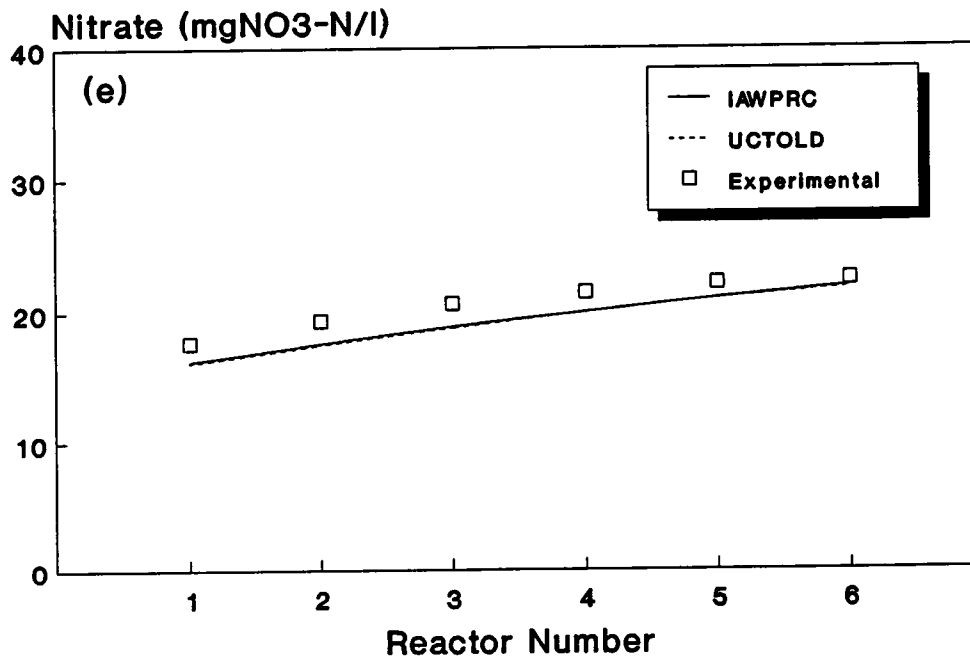
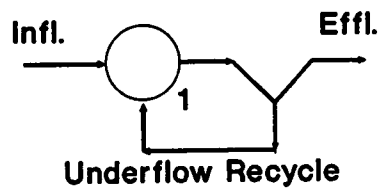


Fig 5.1: Predicted and experimental (a) OUR, (b) COD, (c) VSS, (d) TKN, and (e) nitrate profiles for 6 in-series reactor completely aerobic system under constant flow and load; predictions using UCTOLD and IAWPRC, experimental data from Ekama and Marais (1979).

Cyclic flow and load

Completely mixed single reactor aerobic systems receiving cyclic flow and load influent and operated at sludge ages of 2,5 and 20 days (Ekama and Marais, 1979) were simulated. In both systems, unsettled municipal wastewater from Cape Flats, Cape Town, South Africa, served as influent.



Sludge age 2,5 days

The cyclic flow and load input data were as follows:

***** DIURNAL INPUT PATTERN *****

Record No	Time (h)	Flow(l d-1)	COD (g m-3)	TKN (g m-3)
1	0.0	0.0	515.0	45.8
2	2.0	36.0	515.0	45.8
3	4.0	36.0	515.0	45.8
4	6.0	36.0	515.0	45.8
5	8.0	36.0	515.0	45.8
6	10.0	36.0	515.0	45.8
7	12.0	36.0	515.0	45.8
8	14.0	0.0	515.0	45.8
9	16.0	0.0	515.0	45.8
10	18.0	0.0	515.0	45.8
11	20.0	0.0	515.0	45.8
12	22.0	0.0	515.0	45.8

** Calculated Mean Values :

Flowrate	=	18.0
COD	=	515.0
TKN	=	45.8

Change any values? Y/N....

Wastewater characteristics and configuration and operational data input were as follows:

**** WASTEWATER CHARACTERISTICS ****

Sti(avg) g COD m-3	515.000
Nti(avg) g N m-3	45.800
Fbs g COD g-1 COD	0.190
Fs,us g COD g-1 COD	0.080
Fs,up g COD g-1 COD	0.130
Fn,a g N g-1 N	0.750
Fnob,p g N g-1 N	0.500
Fn,ous g N g-1 N	0.000
Fs,zbh g Zbh COD g-1 COD	0.000
VSS/TSS g VSS g-1 TSS	0.750
Inf Alk mole m-3	10.000

RETURN TO MENU

Fbs = readily biodegradable/biodegradable COD
Fs,us = frac infl COD that is sol unbiodegrad
Fs,up = frac infl COD that is part unbiodegrad
Fn,a = frac infl TKN that is NH3/NH4
Fnob,p = frac organic bio N that is part
Fn,ous = frac infl TKN that is organic unbio sol
Fs,zbh = frac infl COD that is heterotrophs

Hit <Arrows>/<SpaceBar>
to move selection

Hit <Return> to enter
new value for
selected constant

**** PROCESS CONFIGURATION ****

Gp 1. Number of Reactors = 1

Gp 2. Reactor Vols, l		Gp 3. Feed Fraction	Gp 4. Aeration/DO
No. 1:	6.73	1.00	3.0

Do you wish to change any parameters?
Y/N....

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***** PLANT OPERATING PARAMETERS *****
1  SRT {total}          d          =      2.5
2  Process Temperature  degC       =      20.0

Flow rates:
3  Influent flow        l d-1      =      18.0
4  RAS recycle flow     l d-1      =      18.0

```

Do you wish to change any parameters?

Y/N....

Sludge wastage to maintain the sludge age took place continuously during the feed on period, as follows:

```

***** SLUDGE WASTAGE PATTERN *****
Record No    Time (h)    Wastage On    Waste Flow ( l d-1)
1             0.0        No            0.00
2             2.0        Yes           5.38
3             4.0        Yes           5.38
4             6.0        Yes           5.38
5             8.0        Yes           5.38
6            10.0        Yes           5.38
7            12.0        Yes           5.38
8            14.0        No            0.00
9            16.0        No            0.00
10           18.0        No            0.00
11           20.0        No            0.00
12           22.0        No            0.00

** Mean wastage rate =
    2.69 l/d

```

Switch any on/off? Y/N....

The value for $\hat{\mu}_A$ had to be increased to 0,65/d. Also, since in the system pH = 7,0, the autotroph kinetic constants had to be adjusted (WRC, 1984) using Eqs (5.1) and (5.2) to give:

$$\begin{aligned}\hat{\mu}_A &= 0,65 \cdot (2,35)^{(7,0-7,2)} \\ &= 0,548 \text{ /d}\end{aligned}$$

$$\begin{aligned}K_{SA} &= 1,0 \cdot (2,35)^{(7,2-7,0)} \\ &= 1,186 \text{ mgN/l}\end{aligned}$$

The values for all other kinetic and stoichiometric constants were the default values.

The predicted and measured results are compared in Fig 5.2(a,b,c,d and e); good agreement is obtained with both models. This system can be used to determine the influent readily biodegradable COD concentration, see Chapter 6 for a detailed description of the test.

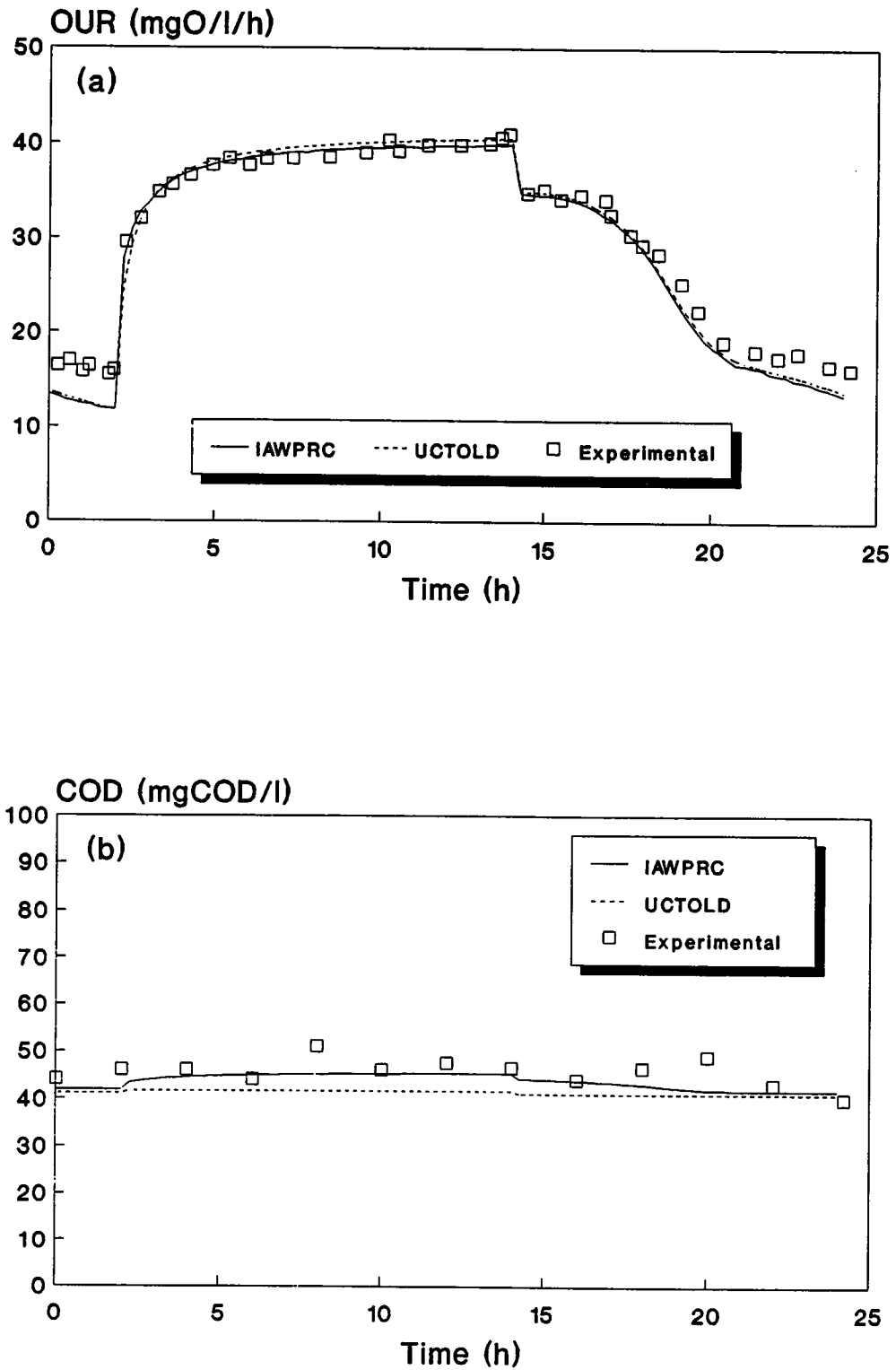


Fig 5.2: Predicted and experimental (a) OUR, (b) COD, (c) VSS, (d) TKN, and (e) nitrate time profiles for single reactor completely aerobic system operated at 2.5d sludge age under cyclic flow and load (feed 12h on, 12h off); predictions using UCTOLD and IAWPRC, experimental data from Ekama and Marais (1979). Continued overleaf.....

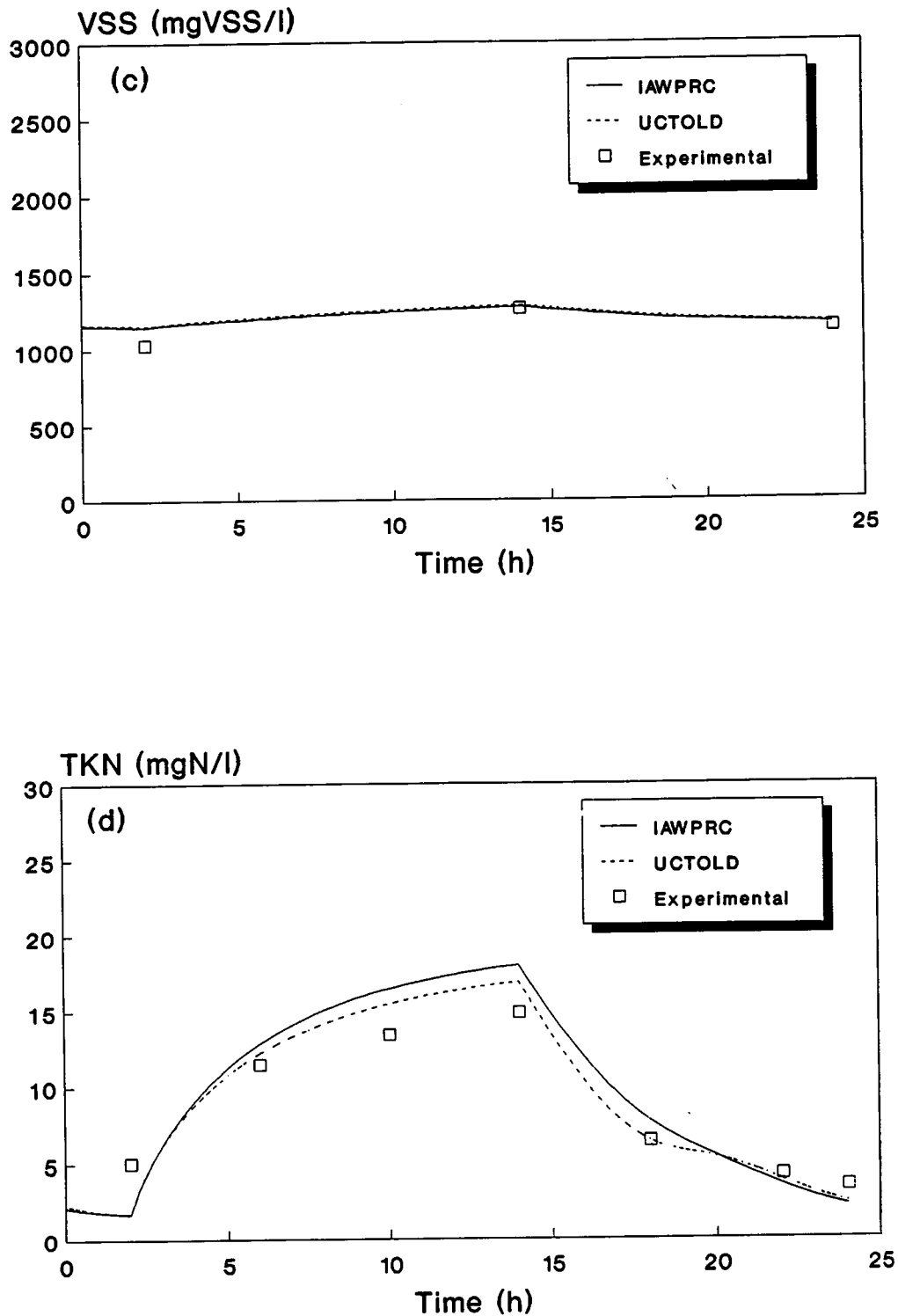


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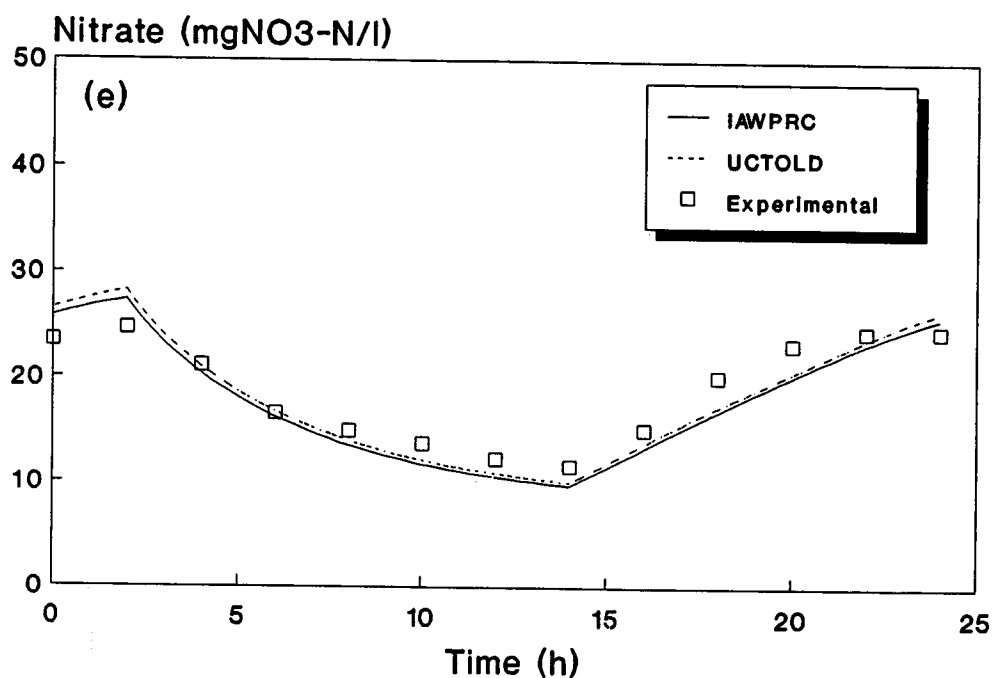


Fig 5.2: Predicted and experimental (a) OUR, (b) COD, (c) VSS, (d) TKN, and (e) nitrate time profiles for single reactor completely aerobic system operated at 2,5d sludge age under cyclic flow and load (feed 12h on, 12h off); predictions using UCTOLD and IAWPRC, experimental data from Ekama and Marais (1979).

Sludge age 20 days

The cyclic flow and load input data were as follows:

***** DIURNAL INPUT PATTERN *****				
Record No	Time (h)	Flow(l d-1)	COD (g m-3)	TKN (g m-3)
1	0.0	0.0	515.0	44.1
2	2.0	28.0	515.0	44.1
3	4.0	28.0	515.0	44.1
4	6.0	28.0	515.0	44.1
5	8.0	28.0	515.0	44.1
6	10.0	28.0	515.0	44.1
7	12.0	28.0	515.0	44.1
8	14.0	0.0	515.0	44.1
9	16.0	0.0	515.0	44.1
10	18.0	0.0	515.0	44.1
11	20.0	0.0	515.0	44.1
12	22.0	0.0	515.0	44.1

** Calculated Mean Values :				
Flowrate	=	14.0		
COD	=	515.0		
TKN	=	44.1		

Change any values? Y/N....

Sewage characteristics and configuration and operational data input were as follows:

**** WASTEWATER CHARACTERISTICS ****		
Sti(avg)	g COD m-3	515.000
Nti(avg)	g N m-3	44.100
Fbs	g COD g-1 COD	0.190
Fs,us	g COD g-1 COD	0.080
Fs,up	g COD g-1 COD	0.130
Fn,a	g N g-1 N	0.750
Fnob,p	g N g-1 N	0.500
Fn,ous	g N g-1 N	0.000
Fs,zbh	g Zbh COD g-1 COD	0.000
VSS/TSS	g VSS g-1 TSS	0.750
Inf Alk	mole m-3	10.000
RETURN TO MENU		

Fbs = readily biodegradable/biodegradable COD
 Fs,us = frac infl COD that is sol unbiodegrad
 Fs,up = frac infl COD that is part unbiodegrad
 Fn,a = frac infl TKN that is NH3/NH4
 Fnob,p = frac organic bio N that is part
 Fn,ous = frac infl TKN that is organic unbio sol
 Fs,zbh = frac infl COD that is heterotrophs

Hit <Arrows>/<SpaceBar>
to move selection

Hit <Return> to enter
new value for
selected constant

**** PROCESS CONFIGURATION ****		Gp 1. Number of Reactors = 1	
Gp 2. Reactor Vols, l		Gp 3. Feed Fraction	Gp 4. Aeration/DO
No. 1:	12.00	1.00	3.0

Do you wish to change any parameters?

Y/N....

***** PLANT OPERATING PARAMETERS *****			
1	SRT {total}	d	= 20.0
2	Process Temperature	degC	= 20.0
Flow rates:			
3	Influent flow	l d-1	= 14.0
4	RAS recycle flow	l d-1	= 14.0

Do you wish to change any parameters?

Y/N....

Sludge wastage to maintain the sludge age took place continuously during the feed on period, as follows:

***** SLUDGE WASTAGE PATTERN *****			
Record No	Time (h)	Wastage On	Waste Flow (l d-1)
1	0.0	No	0.00
2	2.0	Yes	1.20
3	4.0	Yes	1.20
4	6.0	Yes	1.20
5	8.0	Yes	1.20
6	10.0	Yes	1.20
7	12.0	Yes	1.20
8	14.0	No	0.00
9	16.0	No	0.00
10	18.0	No	0.00
11	20.0	No	0.00
12	22.0	No	0.00
** Mean wastage rate =			
0.60 l/d			
Switch any on/off? Y/N....			

The value for $\hat{\mu}_A$ had to be increased to 0,75/d. Also, since in the system pH = 5,5, the autotroph kinetic constants had to be adjusted (WRC, 1984) using Eqs (5.1) and (5.2), to give:

$$\begin{aligned}\hat{\mu}_A &= 0,75 \cdot (2,35)^{(5,5-7,2)} \\ &= 0,175 \text{ /d}\end{aligned}$$

$$\begin{aligned}K_{SA} &= 1,0 \cdot (2,35)^{(7,2-5,5)} \\ &= 4,27 \text{ mgN/l}\end{aligned}$$

The values used in the simulation for all other kinetic and stoichiometric constants were the default values. The predicted and measured results are compared in Figs 5.3(a,b,c,d and e); good agreement is obtained with both models.

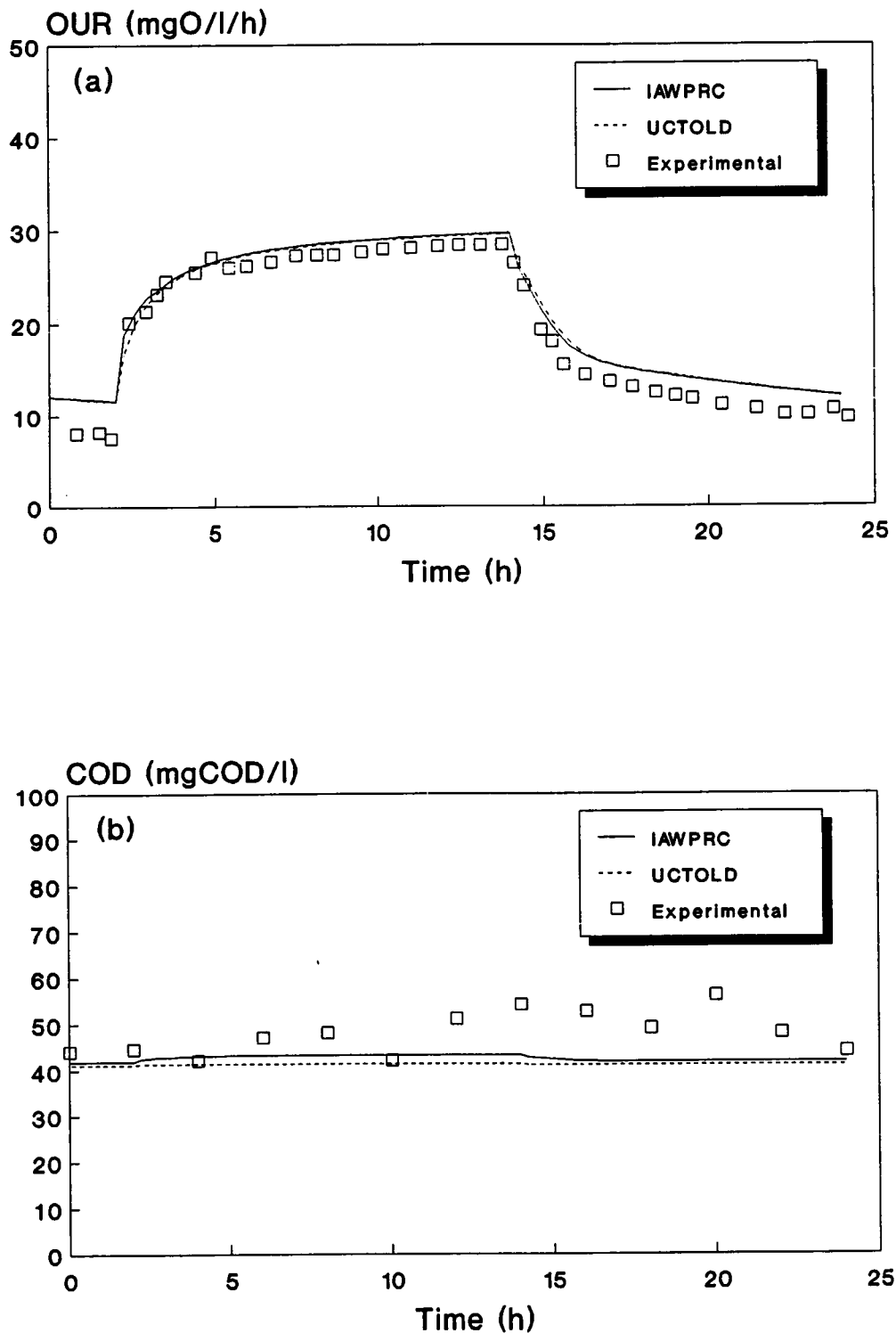


Fig 5.3: Predicted and experimental (a) OUR, (b) COD, (c) VSS, (d) TKN, and (e) nitrate time profiles for single reactor completely aerobic system operated at 20d sludge age under cyclic flow and load (feed 12h on, 12h off); predictions using UCTOLD and IAWPRC, experimental data from Ekama and Marais (1979). Continued overleaf.....

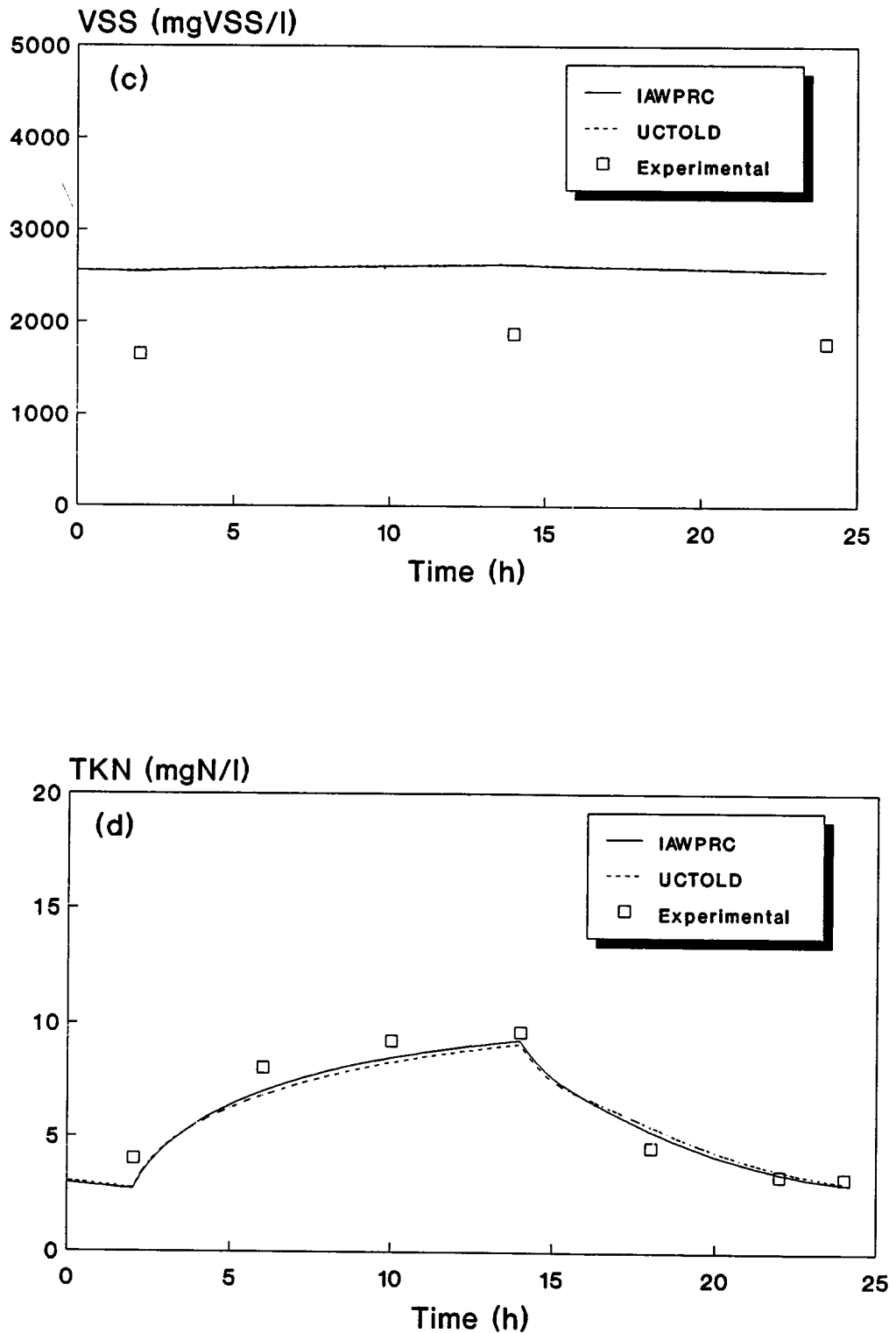


Fig 5.3: Predicted and experimental (a) OUR, (b) COD, (c) VSS, (d) TKN, and (e) nitrate time profiles for single reactor completely aerobic system operated at 20d sludge age under cyclic flow and load (feed 12h on, 12h off); predictions using UCTOLD and IAWPRC, experimental data from Ekama and Marais (1979). Continued overleaf.....

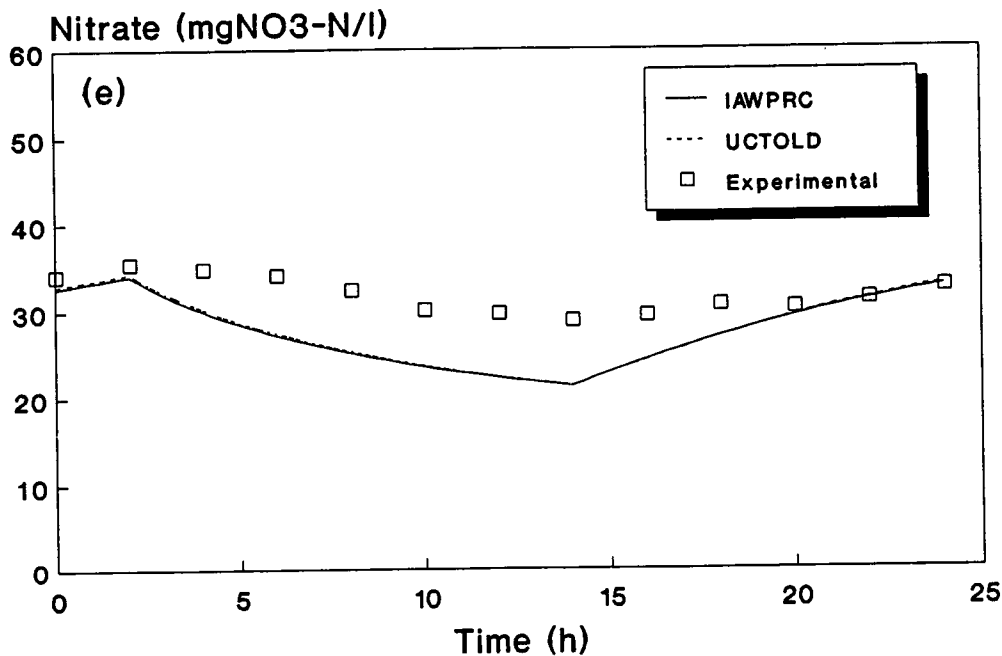


Fig 5.3: Predicted and experimental (a) OUR, (b) COD, (c) VSS, (d) TKN, and (e) nitrate time profiles for single reactor completely aerobic system operated at 20d sludge age under cyclic flow and load (feed 12h on, 12h off); predictions using UCTOLD and IAWPRC, experimental data from Ekama and Marais (1979).

AEROBIC BATCH TESTS

To perform the aerobic batch test, mixed liquor from an operating aerobic or anoxic/aerobic activated sludge system (parent system) is combined with a quantity of the same wastewater that serves as influent to the parent system, the mixture is aerated, and the response of various parameters monitored with time (see Chapter 6 for details on test procedures). These batch tests can be conducted with nitrification present or with nitrification inhibited by addition of thiourea, or a similar compound. Simulations of batch tests with nitrification, and with nitrification inhibited, were undertaken. The simulations were conducted using batch versions of the two models; the batch programs are not included in the simulation package and the simulations are merely to illustrate additional model verification.

In simulation of the batch tests, the initial mixed liquor constituent fractions (active mass, endogenous mass, etc.) were calculated from the parent system parameters (sludge age, influent COD, etc.) using the steady state theory (WRC, 1984; Chapter 6) or the UCTOLD/IAWPRC computer programs. The constituent fractions of the wastewater had been determined separately (for procedures see

Chapter 6). The combination of mixed liquor and wastewater constituent fractions in the batch test gives the initial starting condition (i.e. at time = 0) and serves as input to the batch program.

From the batch test simulations three aspects of importance became apparent:

- (1) To obtain the same simulated oxygen utilization rates (OUR) using the IAWPRC Task Group and UCT models, the input value for the heterotroph maximum specific growth rate on RBCOD ($\hat{\mu}_H$) differs. This is so because in the IAWPRC Task Group model growth occurs only with RBCOD, and the OUR associated with this growth (plus any nitrification OUR) gives rise to the observed OUR; in the UCT model growth occurs both on RBCOD and SBCOD and the growth processes on these two substrate types are independent and, consequently, the summation of the OUR associated with the respective growth processes on RBCOD and SBCOD (plus any nitrification OUR) gives rise to the observed OUR, see **Chapters 2 and 6**.
- (2) In both models variation in the reactor configuration of, and load pattern on, the parent system from which the mixed liquor is taken for the batch test, can give rise to markedly different $\hat{\mu}_H$ values. Such differences in $\hat{\mu}_H$ were noted previously in investigations into sludge bulking and were linked to stimulation of a "selector effect" (Jenkins *et al.*, 1984; Still *et al.*, 1986; Gabb *et al.*, 1989). In contrast an important conclusion from the batch simulations is that the maximum specific rate of SBCOD hydrolysis appears to be independent of reactor configuration and load pattern, and can be accepted as constant.
- (3) In both models it was necessary to change the maximum specific growth rate for the autotrophs ($\hat{\mu}_A$) when simulating batch tests with different influents and/or batch tests with mixed liquor drawn from different systems in order to predict correctly the nitrate-time profiles. In some of the simulations described earlier, the value for $\hat{\mu}_A$ also had to be changed from the default value and from system to system. This behaviour had been noted previously where it was concluded that the value for $\hat{\mu}_A$ is dependent on the specific wastewater (WRC, 1984) and on the reactor configuration and load pattern (Still *et al.*, 1986). (For more detailed discussion, see **Chapter 6**.)

The variability in $\hat{\mu}_H$ and $\hat{\mu}_A$ necessitates that values for these constants be determined for each batch test, by using the procedures set out in **Chapter 6**. Accordingly, values for these two constants will be given where appropriate for each simulation described below.

Batch test with nitrification

One example of an aerobic batch test with nitrification was selected from a number of similar tests. In this test, mixed liquor was drawn from the parent system (anoxic/aerobic fill and draw, sludge age = 20d, reactor volume = 10ℓ, feed = 7500 mgCOD/d unsettled municipal wastewater) and mixed with unsettled municipal wastewater from Mitchell's Plain, South Africa to give the initial starting conditions as follows:

For UCTOLD

INITIAL CONCENTRATIONS

1 Zbh (hetero.)	=	460.00 g COD m ⁻³
2 Ze (endog.)	=	442.00 g COD m ⁻³
3 Zba (autotrophs)	=	25.40 g COD m ⁻³
4 Sads (adsorb COD)	=	5.00 g COD m ⁻³
5 Serbm (enmesh COD)	=	173.96 g COD m ⁻³
6 Zi (prt unb COD)	=	17.44 g COD m ⁻³
7 Znd (prt bio N)	=	1.77 g N m ⁻³
8 Sbs (sol bio COD)	=	9.16 g COD m ⁻³
9 Na (ammonia N)	=	14.17 g N m ⁻³
10 Nobp (sol org N)	=	1.77 g N m ⁻³
11 No3 (nitrate N)	=	9.40 g N m ⁻³
12 Alkalinity	=	7.50 mole m ⁻³
13 Sus (sol unb COD)	=	17.44 g COD m ⁻³

Do you wish to change any values?...Y/N

For IAWPRC

INITIAL CONCENTRATIONS

1 Zbh (hetero.)	=	460.00 g COD m ⁻³
2 Ze (endog.)	=	442.00 g COD m ⁻³
3 Zba (autotrophs)	=	25.40 g COD m ⁻³
4 Serbm (enmesh COD)	=	173.96 g COD m ⁻³
5 Zi (prt unb COD)	=	17.44 g COD m ⁻³
6 Nobp (prt bio N)	=	1.77 g N m ⁻³
7 Sbs (sol bio COD)	=	9.16 g COD m ⁻³
8 Na (ammonia N)	=	14.17 g N m ⁻³
9 Nobs (sol org N)	=	4.56 g N m ⁻³
10 No3 (nitrate N)	=	9.40 g N m ⁻³
11 Alkalinity	=	7.50 mole m ⁻³
12 Sus (sol unb COD)	=	17.44 g COD m ⁻³

Do you wish to change any values?...Y/N

For the simulations, $\hat{\mu}_H = 5$ and $4,3/d$ for IAWPRC and UCTOLD respectively, and $\hat{\mu}_A = 0,29/d$ for both models. All other kinetic and stoichiometric constants were the default values. The predicted and measured OUR and concentration time profiles are compared in Fig 5.4(a,b and c); good agreement is obtained.

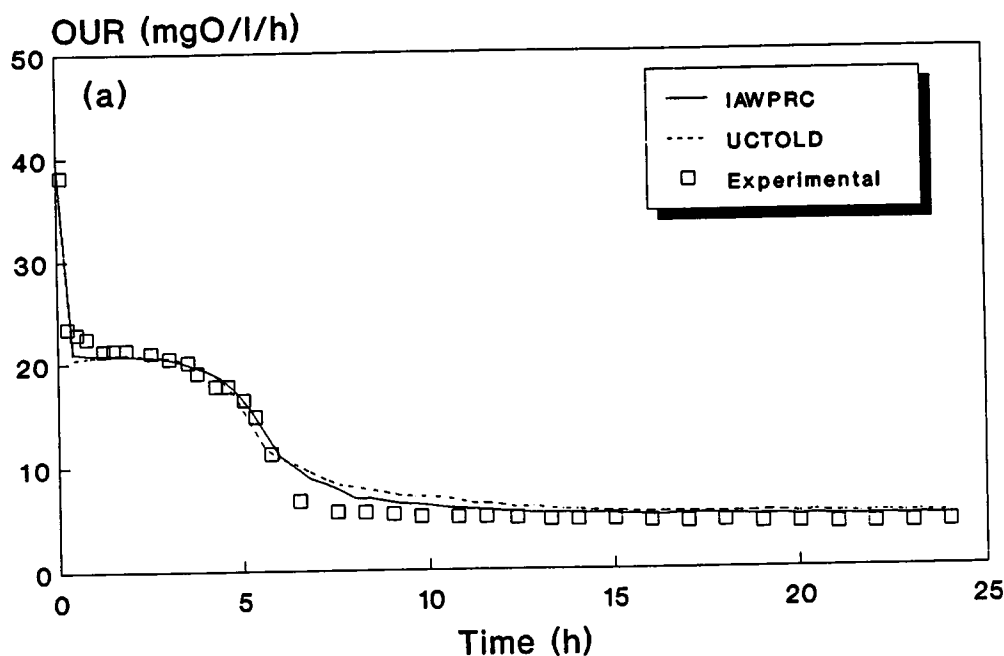


Fig 5.4: Predicted and experimental (a) OUR, (b) nitrate, and (c) TKN time profiles for aerobic batch test with nitrification; predictions using batch versions of UCTOLD and IAWPRC programs with default value for SBCOD hydrolysis half saturation constant (K_{sp} UCTOLD; K_X IAWPRC). Continued overleaf.....

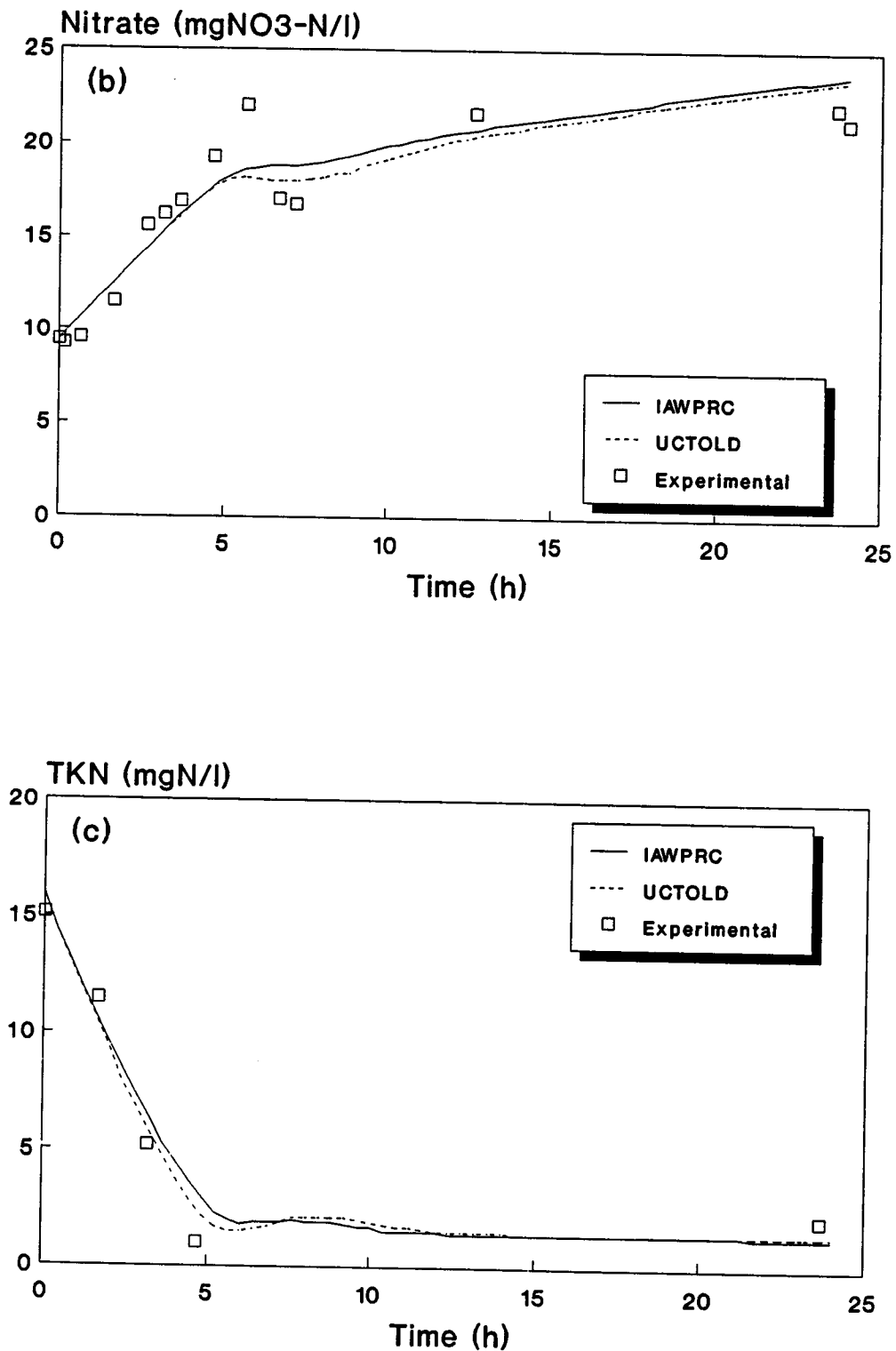


Fig 5.4: Predicted and experimental (a) OUR, (b) nitrate, and (c) TKN time profiles for aerobic batch test with nitrification; predictions using batch versions of UCTOLD and IAWPRC programs with default value for SBCOD hydrolysis half saturation constant (K_{SP} UCTOLD; K_X IAWPRC).

Batch tests with nitrification inhibition

In this series of aerobic batch tests nitrification was inhibited by addition of 20 mg thiourea per litre of batch volume. In an example of one such test, mixed liquor was drawn from the parent system (completely aerobic system with selector reactor, sludge age = 20d, total reactor volume = 15ℓ – selector reactor volume = 0,3ℓ and main reactor volume = 14,7ℓ, feed = 30 ℓ/d at 350 mgCOD/ℓ) and mixed with settled municipal wastewater from Mitchell's Plain, South Africa to give initial conditions as follows:

For UCTOLD

INITIAL CONCENTRATIONS *****

1 Zbh (hetero.)	=	379.00 g COD m ⁻³
2 Ze (endog.)	=	353.00 g COD m ⁻³
3 Zba (autotrophs)	=	0.00 g COD m ⁻³
4 Sads (adsorb COD)	=	5.00 g COD m ⁻³
5 Seru (ermesh COD)	=	220.33 g COD m ⁻³
6 Zi (prt unb COD)	=	8.94 g COD m ⁻³
7 Znd (prt bio N)	=	2.25 g N m ⁻³
8 Sbs (sol bio COD)	=	35.87 g COD m ⁻³
9 Na (ammonia N)	=	24.90 g N m ⁻³
10 Nobp (sol org N)	=	2.25 g N m ⁻³
11 No3 (nitrate N)	=	10.00 g N m ⁻³
12 Alkalinity	=	7.50 mole m ⁻³
13 Sus (sol unb COD)	=	32.77 g COD m ⁻³

For IAWPRC

INITIAL CONCENTRATIONS *****

1 Zbh (hetero.)	=	379.00 g COD m ⁻³
2 Ze (endog.)	=	353.00 g COD m ⁻³
3 Zba (autotrophs)	=	0.00 g COD m ⁻³
4 Seru (ermesh COD)	=	220.33 g COD m ⁻³
5 Zi (prt unb COD)	=	8.94 g COD m ⁻³
6 Nobp (prt bio N)	=	2.25 g N m ⁻³
7 Sbs (sol bio COD)	=	35.87 g COD m ⁻³
8 Na (ammonia N)	=	24.90 g N m ⁻³
9 Nobs (sol org N)	=	2.25 g N m ⁻³
10 No3 (nitrate N)	=	10.00 g N m ⁻³
11 Alkalinity	=	7.50 mole m ⁻³
12 Sus (sol unb COD)	=	32.77 g COD m ⁻³

Do you wish to change any values?...Y/N

Do you wish to change any values?...Y/N

In the simulations $\hat{\mu}_H = 3,0$ and $1,6/d$ for IAWPRC and UCTOLD respectively, and $\hat{\mu}_A = 0/d$ for both models due to nitrification inhibition. All other kinetic and stoichiometric constants were assigned the default values. The predicted and measured OUR-time profiles are compared in Fig 5.5; gross differences between the predicted and measured OUR responses are evident. It was hypothesized that the thiourea exerts some inhibitory influence on the heterotrophic population. This inhibition appeared to be most evident in the rate of hydrolysis/utilization of SBCOD. Accordingly, a study was made on the effect of changing the rate constants associated with SBCOD hydrolysis/utilization on the model predictions. This showed that the value for the maximum specific hydrolysis rate ($K_H = 2,0/d$, IAWPRC; $K_{MP} = 1,35/d$ UCTOLD) could be retained, but the hydrolysis half saturation constant (K_X , IAWPRC; K_{SP} , UCTOLD) needed to be increased, from 0,027 to 0,15 mgCOD/mgCOD, for both models. With only this change the OUR responses in the batch tests with nitrification inhibition were closely simulated by both models, see Fig 5.6. Using the increased value for the hydrolysis half saturation constant in the models, simulations of the spectrum of system configurations and operating conditions gave poor fits of the predicted responses to the measured response; as an example, the batch test with nitrification was simulated using the increased hydrolysis half saturation constant, see Fig 5.7. Evidently thiourea exerts a significant influence on the hydrolysis kinetic rate, and that this can be accommodated in the structure of the two models by increasing the magnitude of one constant, the half saturation constant for hydrolysis. Inhibition of heterotrophs by thiourea had been noted previously at the University of Cape Town, in investigations into biological phosphorus removal (Wentzel *et al.*, 1988) and has been substantiated subsequently where it was observed that addition of 120 mg thiourea/ℓ batch volume (6 times normal concentration) completely inhibited heterotrophic activity.

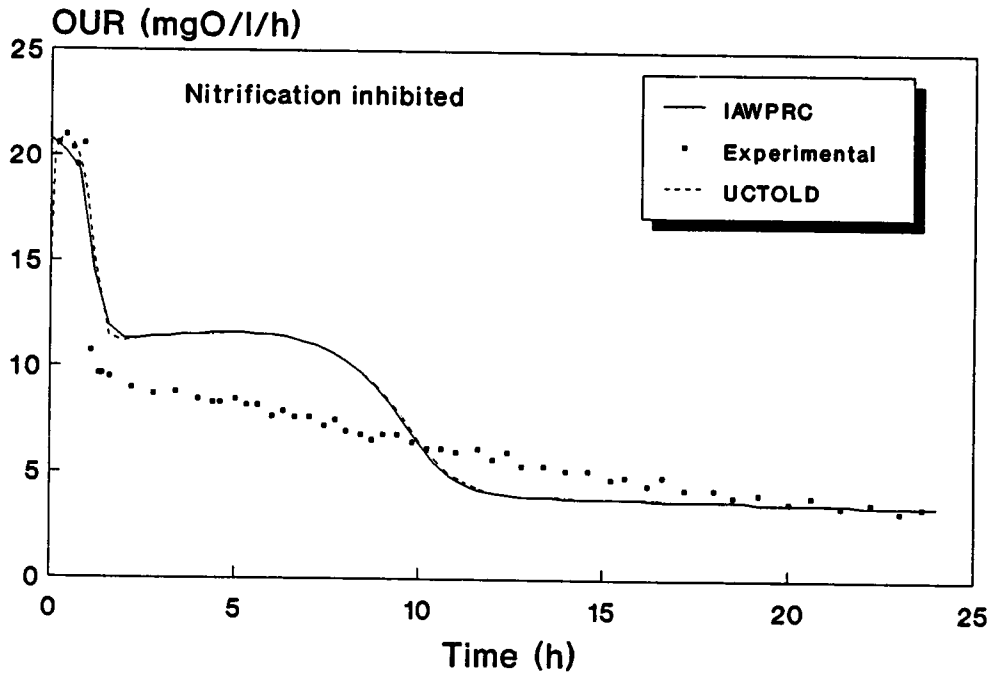


Fig 5.5: Predicted and experimental OUR-time profiles for aerobic batch test with nitrification inhibited by addition of 20 mg thiourea/l batch volume; predictions using batch versions of UCTOLD and IAWPRC programs with default value for SBCOD hydrolysis half saturation constant (K_{sp} UCTOLD; K_X IAWPRC).

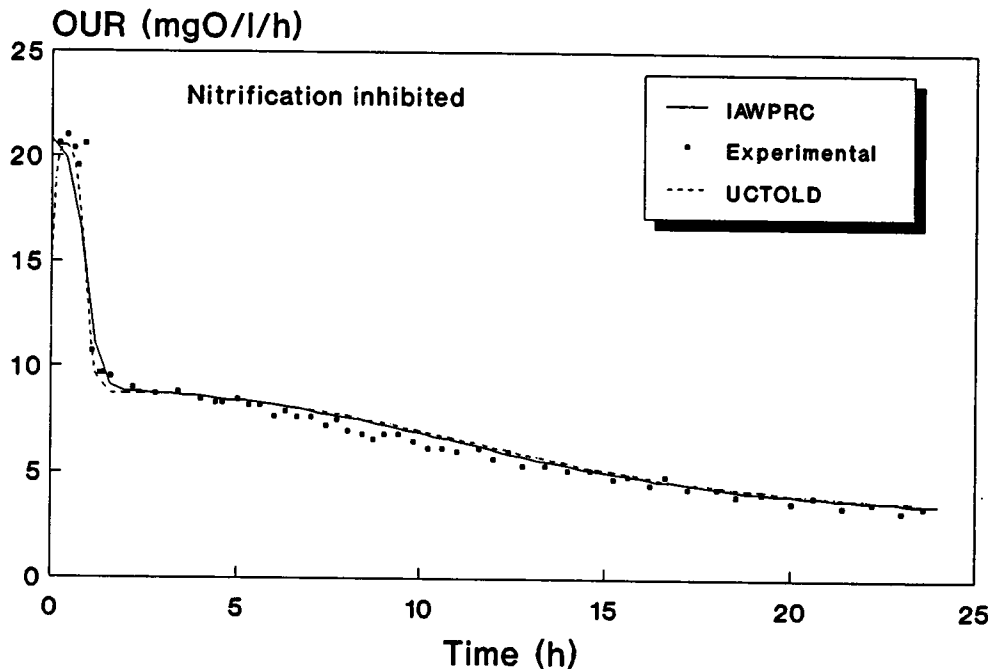


Fig 5.6: Predicted and experimental OUR-time profiles for aerobic batch test with nitrification inhibited by addition of 20 mg thiourea/l batch volume; predictions using batch versions of UCTOLD and IAWPRC programs with the SBCOD hydrolysis half saturation constant (K_{sp} UCTOLD; K_X IAWPRC) increased from the default value of 0,027 to 0,15 mgCOD/mgCOD (see Fig 5.5 for simulations with the default value for this constant).

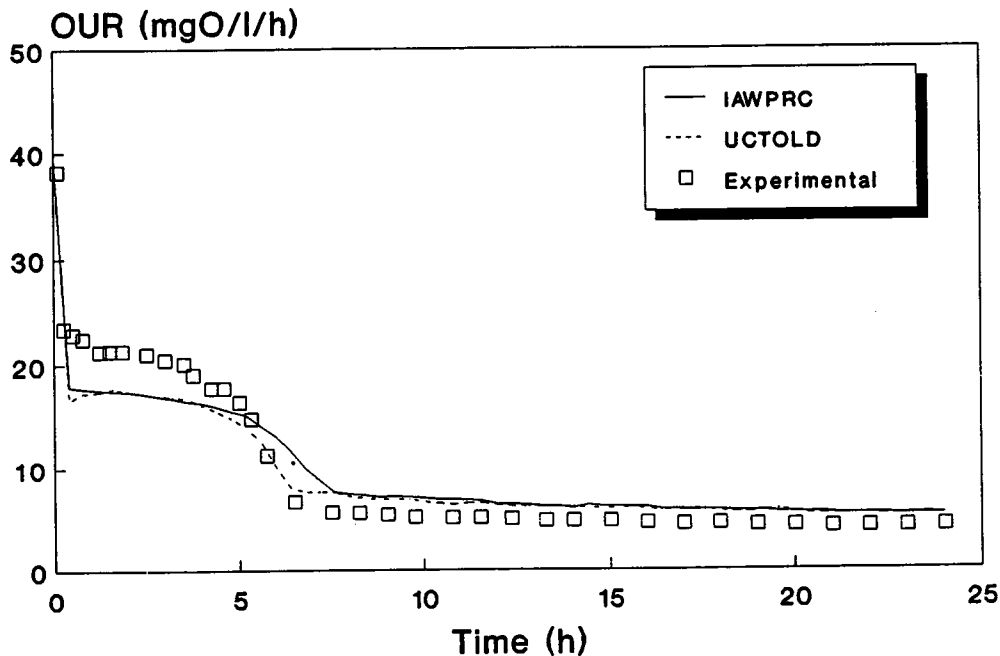


Fig 5.7: Predicted and experimental OUR-time profiles for aerobic batch test with nitrification; predictions using batch versions of UCTOLD and IAWPRC programs with the SBCOD hydrolysis half saturation constant increased from the default value of 0,027 to 0,015 mgCOD/mgCOD (see Fig 5.4 for simulations with the default value for this constant).

ANOXIC/AEROBIC SYSTEMS – PLUGFLOW ANOXIC REACTORS

In nitrification/denitrification systems, Stern and Marais (1974) had shown that in plugflow anoxic reactors upstream of the aerobic reactor (primary anoxic reactor) denitrification took place in two linear phases, a rapid first phase which persisted for a short period then terminated and a second slow phase which continued for the retention time in the reactor. In plugflow anoxic reactors downstream of the aerobic reactor (secondary anoxic reactor), only one linear denitrification phase was operative, at a slow rate of about two thirds that of the slow second rate in the primary anoxic reactor. Stern and Marais concluded that the three denitrification phases can be expressed by:

$$\Delta N = K X_{BH} \Delta R$$

where K = specific denitrification rate constant per unit active mass per unit time
 X_{BH} = heterotroph biological active mass concentration
 ΔR = actual reaction time
 ΔN = nitrate reduction per unit volume

From the experimental data available, van Haandel *et al.* (1981) evaluated the mean values for the specific denitrification rate constants, K , for the three denitrification phases (K_1 , K_2 and K_3 respectively) for various sludge ages and temperatures; values for K_2 and K_3 are listed in Table 5.2.

The denitrification behaviour in plugflow primary and secondary anoxic reactors was simulated for a range of sludge ages and temperatures using IAWPRC and UCTOLD with the default values for the kinetic and stoichiometric constants. In the simulations, plugflow conditions were approximated by incorporating a series of small completely mixed reactors. As an example, input data for the simulation of the K_2 specific denitrification rate constant at 20°C and 20 days sludge age were as follows:

**** WASTEWATER CHARACTERISTICS ****		
Sti(avg)	g COD m-3	500.000
Nti(avg)	g N m-3	100.000
Fbs	g COD g-1 COD	0.200
Fs,us	g COD g-1 COD	0.050
Fs,up	g COD g-1 COD	0.130
Fn,a	g N g-1 N	0.750
Fnob,p	g N g-1 N	0.500
Fn,ous	g N g-1 N	0.000
Fs,zbh	g Zbh COD g-1 COD	0.000
VSS/TSS	g VSS g-1 TSS	0.750
Inf Alk	mole m-3	10.000
RETURN TO MENU		

Fbs = readily biodegradable/biodegradable COD
 Fs,us = frac infl COD that is sol unbiodegrad
 Fs,up = frac infl COD that is part unbiodegrad
 Fn,a = frac infl TKN that is NH3/NH4
 Fnob,p = frac organic bio N that is part
 Fn,ous = frac infl TKN that is organic unbio sol
 Fs,zbh = frac infl COD that is heterotrophs

Hit <Arrows>/<SpaceBar>
to move selection

Hit <Return> to enter
new value for
selected constant

**** PROCESS CONFIGURATION ****			Gp 1. Number of Reactors = 6
Gp 2. Reactor Vols, l		Gp 3. Feed Fraction	Gp 4. Aeration/DO
No. 1:	1.00	1.00	Unaerated
No. 2:	2.00		Unaerated
No. 3:	2.00		Unaerated
No. 4:	2.00		Unaerated
No. 5:	2.00		Unaerated
No. 6:	10.00		3.0
Gp 5. Recycles: Use to include/remove mixed liquor recycles.			
RAS recycle to Reactor No.1			

Do you wish to change any parameters?

Y/N....

***** PLANT OPERATING PARAMETERS *****				
1	SRT (total)	d	=	20.0
2	Process Temperature	degC	=	20.0
Flow rates:				
3	Influent flow	l d-1	=	15.0
4	RAS recycle flow	l d-1	=	15.0

Do you wish to change any parameters?
.Y/N....

In the simulations, the value for the constant defining the specific denitrification rate for the rapid first phase denitrification in the primary anoxic reactor (K_1) was not evaluated; this rate is linked to the utilization of RBCOD and, as shown in the previous section, the rate is dependent on the system configuration and load pattern through variation in $\hat{\mu}_H$.

The specific denitrification rates, K_2 and K_3 , predicted in the simulations for a range of sludge ages and temperatures are listed in Table 5.2, together with the experimentally observed values; good agreement is obtained with both models.

Table 5.2: Experimentally observed and simulated specific denitrification rate constants K_2 and K_3 for various temperatures and sludge ages.

Temp. (° C)	Sludge Age (d)	Specific denitrification rate constant (mgN/mg X_{BH} /d)					
		Experimental		UCTOLD		IAWPRC	
		K_2	K_3	K_2	K_3	K_2	K_3
15	20	0,0464	0,0461	0,0486	0,0464	0,0495	0,0465
20	10	0,0681	0,0535	0,0703	0,0595	0,0709	0,0572
	20	0,0681	0,0535	0,0681	0,0581	0,0681	0,0568
	30	0,0681	0,0535	0,0681	0,0554	0,0681	0,0558

ANOXIC/AEROBIC SYSTEMS - BATCH DIGESTION

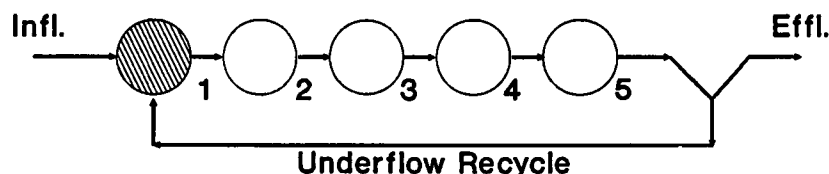
In anoxic/aerobic batch digestion of waste activated sludge, Warner *et al.* (1984) found that the denitrification rate observed (K_4) is lower than that in the secondary anoxic reactor (K_3). From simulations of this system using the UCT model with the default constants in a specialized digester program, Warner *et al.* concluded that the model predicts accurately both the aerobic and anoxic/aerobic digestion behaviour under both steady state and dynamic loading conditions, i.e. the model predicts correctly the lower K_4 denitrification behaviour in anoxic/aerobic batch digestion. The two computer programs included in the simulation package (UCTOLD and IAWPRC) are not set up for the batch digestion system and therefore cannot be used to simulate it; the reader is referred to Warner *et al.* (1983) for details on development of specialized digester programs.

ANOXIC/AEROBIC SYSTEMS - COMPLETELY MIXED REACTORS

Anoxic/aerobic completely mixed in-series reactor systems under constant and cyclic flow and load conditions were simulated.

Constant flow and load

The experimental data were obtained from a pilot-scale investigation at Daspoort Sewage Works, Pretoria, South Africa (van Haandel *et al.*, 1981). The influent was settled municipal wastewater and the reactor configuration was an anoxic reactor in series with four aerobic reactors.



Data input to the program was as follows:

**** WASTEWATER CHARACTERISTICS ****		
Sti(avg)	g COD m-3	477.000
Nti(avg)	g N m-3	45.100
Fbs	g COD g-1 COD	0.240
Fs,us	g COD g-1 COD	0.040
Fs,up	g COD g-1 COD	0.040
Fn,a	g N g-1 N	0.830
Fnob,p	g N g-1 N	0.500
Fn,ous	g N g-1 N	0.000
Fs,zbh	g Zbh COD g-1 COD	0.000
VSS/TSS	g VSS g-1 TSS	0.830
Inf Alk	mole m-3	10.000
RETURN TO MENU		

Fbs = readily biodegradable/biodegradable COD
 Fs,us = frac infl COD that is sol unbiodegrad
 Fs,up = frac infl COD that is part unbiodegrad
 Fn,a = frac infl TKN that is NH₃/NH₄
 Fnob,p = frac organic bio N that is part
 Fn,ous = frac infl TKN that is organic unbio sol
 Fs,zbh = frac infl COD that is heterotrophs

Hit <Arrows>/<SpaceBar>
to move selection

Hit <Return> to enter
new value for
selected constant

**** PROCESS CONFIGURATION ****				Gp 1. Number of Reactors = 5
Gp 2. Reactor Vols,m3		Gp 3. Feed Fraction	Gp 4. Aeration/DO	
No. 1:	5.00	1.00	Un aerated	
No. 2:	5.00		3.0	
No. 3:	5.00		3.0	
No. 4:	5.00		3.0	
No. 5:	5.00		3.0	
Gp 5. Recycles: Use to include/remove mixed liquor recycles.				
RAS recycle to Reactor No.1				

Do you wish to change any parameters?
 Y/N....

***** PLANT OPERATING PARAMETERS *****				
1	SRT (total)	d	=	18.0
2	Process Temperature	degC	=	21.6
Flow rates:				
3	Influent flow	m3 d-1	=	40.0
4	RAS recycle flow	m3 d-1	=	120.0

Do you wish to change any parameters?
 Y/N....

All values for the kinetic and stoichiometric constants were the default values except the maximum specific growth rate for the autotrophs ($\hat{\mu}_A$) which was 0,31/d.

The predicted and measured results are compared in Fig 5.8(a,b,c,d,e and f); good agreement is obtained with both models for all the parameters in each reactor.

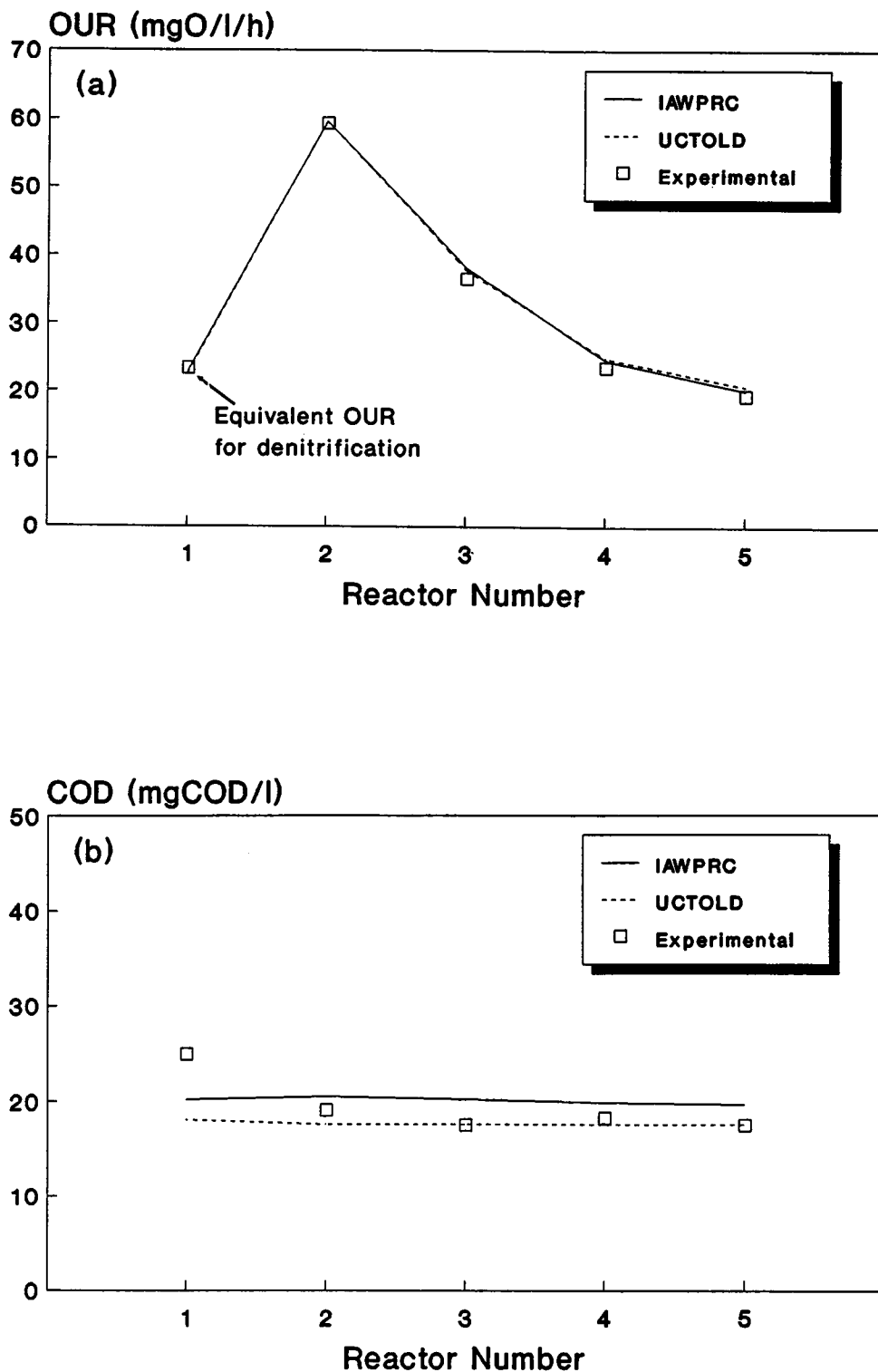


Fig 5.8: Predicted and experimental (a) OUR, (b) COD, (c) VSS, (d) TKN, (e) ammonia, and (f) nitrate profiles for in-series anoxic/4 aerobic reactor system under constant flow and load; predictions using UCTOLD and IAWPRC, data from van Haandel *et al.* (1981). Continued overleaf.....

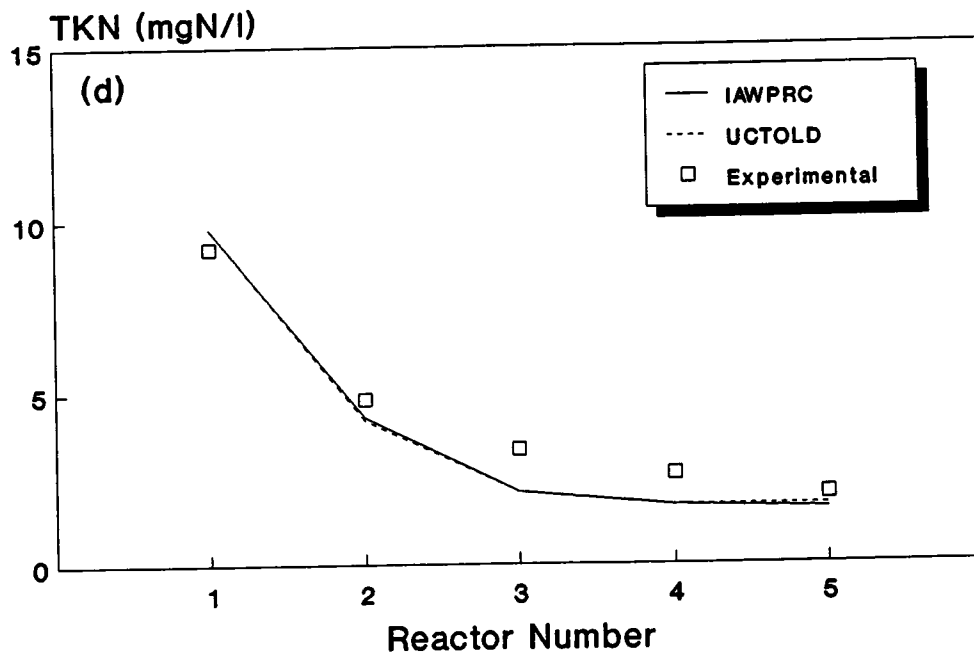
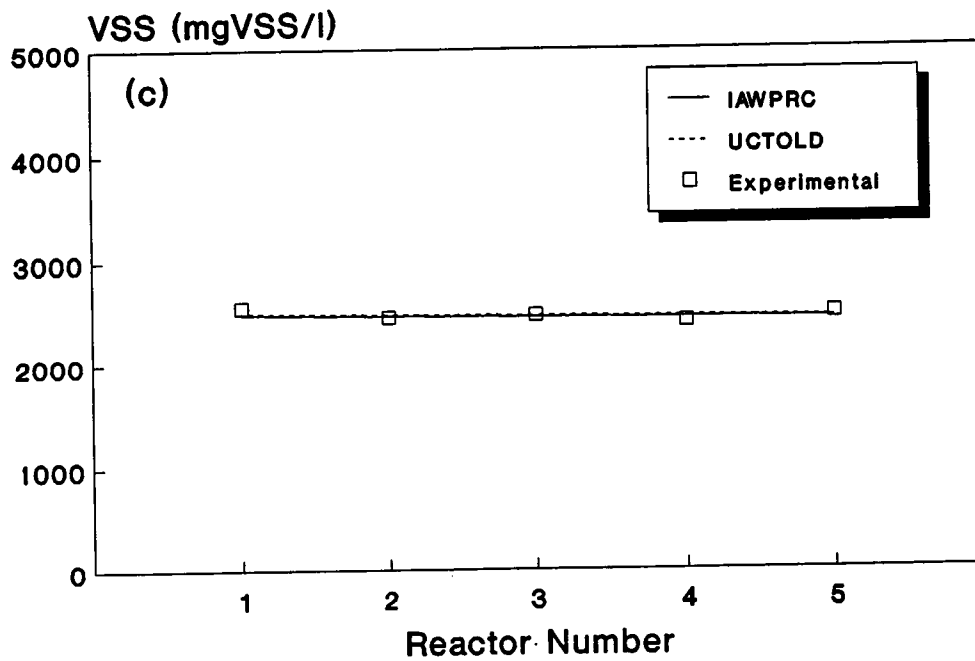


Fig 5.8: Predicted and experimental (a) OUR, (b) COD, (c) VSS, (d) TKN, (e) ammonia, and (f) nitrate profiles for in-series anoxic/4 aerobic reactor system under constant flow and load; predictions using UCTOLD and IAWPRC, data from van Haandel *et al.* (1981). Continued overleaf.....

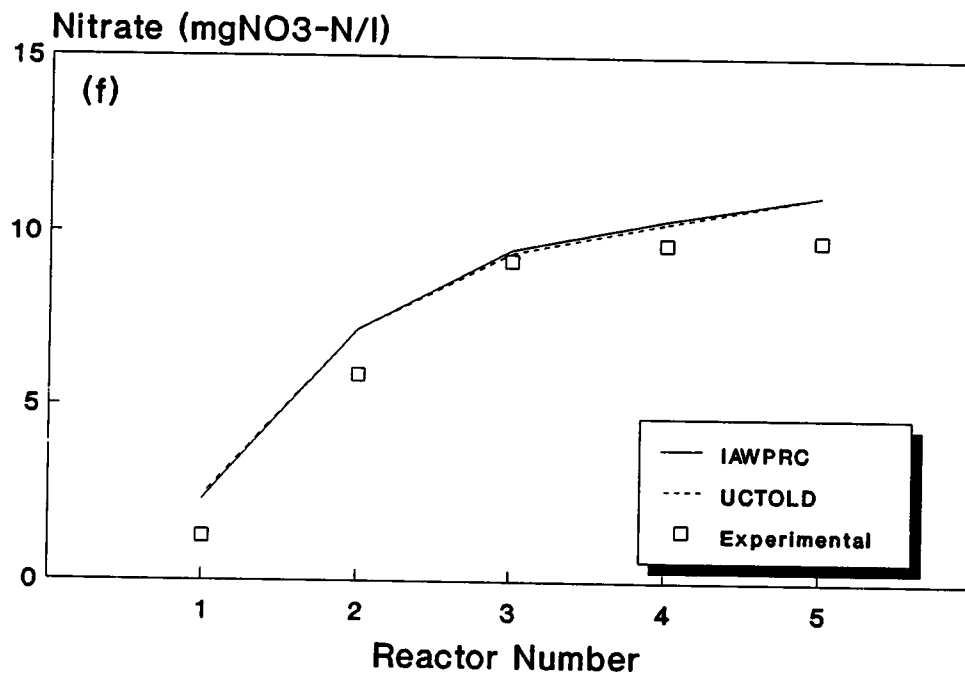
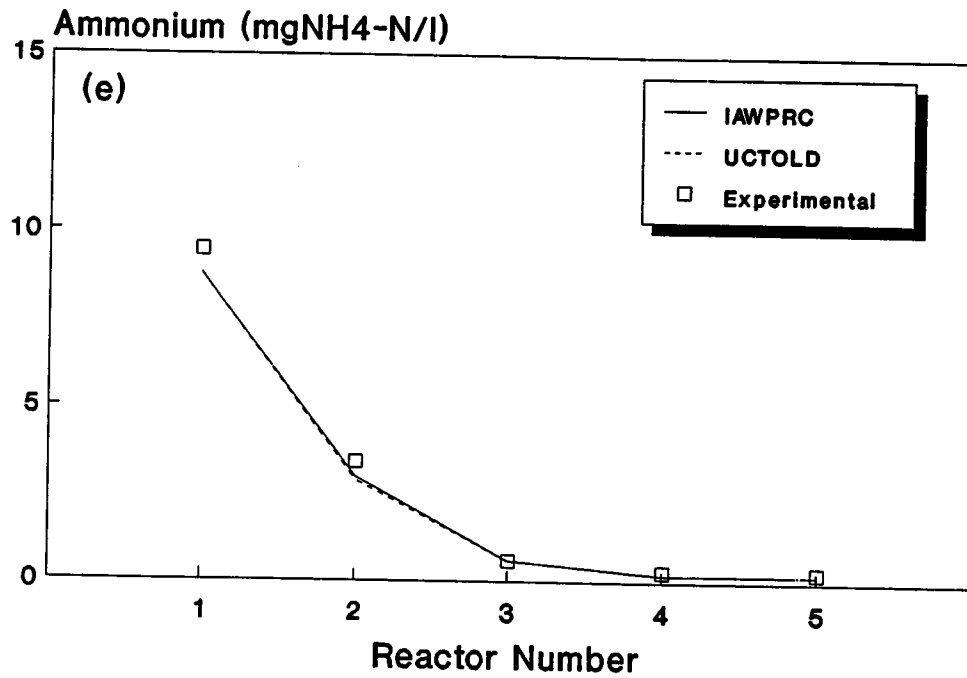
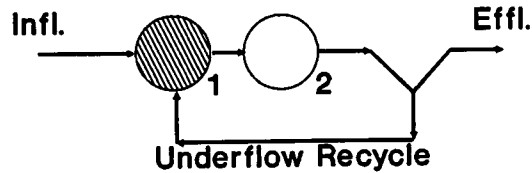


Fig 5.8: Predicted and experimental (a) OUR, (b) COD, (c) VSS, (d) TKN, (e) ammonia, and (f) nitrate profiles for in-series anoxic/4 aerobic reactor system under constant flow and load; predictions using UCTOLD and IAWPRC, data from van Haandel *et al.* (1981).

Cyclic flow and load

At laboratory-scale, Wilson and Marais (1976) tested a two in-series reactor anoxic/aerobic system under cyclic flow and load using settled municipal wastewater from Athlone, Cape Town, South Africa as influent.



The cyclic flow and load input data were as follows:

***** DIURNAL INPUT PATTERN *****				
Record No	Time (h)	Flow(l d-1)	COD (g m-3)	TKN (g m-3)
1	0.0	0.0	570.0	54.0
2	2.0	30.0	570.0	54.0
3	4.0	30.0	570.0	54.0
4	6.0	30.0	570.0	54.0
5	8.0	30.0	570.0	54.0
6	10.0	30.0	570.0	54.0
7	12.0	30.0	570.0	54.0
8	14.0	0.0	570.0	54.0
9	16.0	0.0	570.0	54.0
10	18.0	0.0	570.0	54.0
11	20.0	0.0	570.0	54.0
12	22.0	0.0	570.0	54.0

** Calculated Mean Values :				
Flowrate	=	15.0		
COD	=	570.0		
TKN	=	54.0		

Change any values? Y/N....

Wastewater characteristics and configuration and operational data input were as follows:

**** WASTEWATER CHARACTERISTICS ****	
Sti(avg) g COD m-3	570.000
Nti(avg) g N m-3	54.000
Fbs g COD g-1 COD	0.150
Fs,us g COD g-1 COD	0.200
Fs,up g COD g-1 COD	0.040
Fn,a g N g-1 N	0.830
Fnob,p g N g-1 N	0.500
Fn,ous g N g-1 N	0.040
Fs,zbh g Zbh COD g-1 COD	0.000
VSS/TSS g VSS g-1 TSS	0.830
Inf Alk mole m-3	10.000
RETURN TO MENU	

Fbs = readily biodegradable/biodegradable COD Fs,us = frac infl COD that is sol unbiodegrad Fs,up = frac infl COD that is part unbiodegrad Fn,a = frac infl TKN that is NH3/NH4 Fnob,p = frac organic bio N that is part Fn,ous = frac infl TKN that is organic unbio sol Fs,zbh = frac infl COD that is heterotrophs	Hit <Arrows>/<SpaceBar> to move selection Hit <Return> to enter new value for selected constant
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**** PROCESS CONFIGURATION ****				Gp 1. Number of Reactors = 2
Gp 2. Reactor Vols, l		Gp 3. Feed Fraction	Gp 4. Aeration/DO	
No. 1:	1.00	1.00	Un aerated	
No. 2:	6.40		3.0	
Gp 5. Recycles: Use to include/remove mixed liquor recycles.				
RAS recycle to Reactor No.1				

Do you wish to change any parameters?
Y/N....

***** PLANT OPERATING PARAMETERS *****				
1	SRT {total}	d	=	20.0
2	Process Temperature	degC	=	20.0
Flow rates:				
3	Influent flow	l d-1	=	15.0
4	RAS recycle flow	l d-1	=	22.5

Do you wish to change any parameters?
Y/N....

Sludge wastage to maintain the sludge age took place continuously during the feed on period, as follows:

***** SLUDGE WASTAGE PATTERN *****			
Record No	Time (h)	Wastage On	Waste Flow (l d-1)
1	0.0	No	0.00
2	2.0	Yes	0.74
3	4.0	Yes	0.74
4	6.0	Yes	0.74
5	8.0	Yes	0.74
6	10.0	Yes	0.74
7	12.0	Yes	0.74
8	14.0	No	0.00
9	16.0	No	0.00
10	18.0	No	0.00
11	20.0	No	0.00
12	22.0	No	0.00
** Mean wastage rate =			
0.37 l/d			
Switch any on/off? Y/N....			

Values for all the kinetic and stoichiometric constants were the default values, except for $\hat{\mu}_A$ which was 0.17/d.

The predicted and measured results are compared in Fig 5.9(a,b,c,d and e); good agreement is obtained.

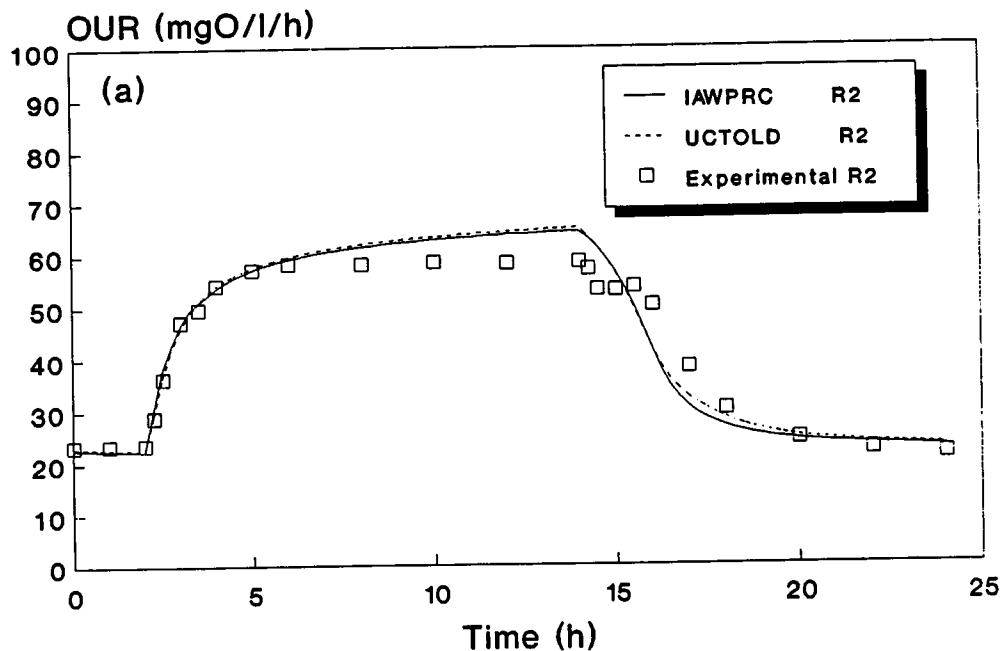


Fig 5.9: Predicted and experimental (a) OUR, (b) COD, (c) VSS, (d) TKN, and (e) nitrate time profiles for in-series anoxic (R1)/aerobic (R2) reactor system under cyclic flow and load (feed 12h on, 12h off); predictions using UCTOLD and IAWPRC, data from Wilson and Marais (1976). Continued overleaf.....

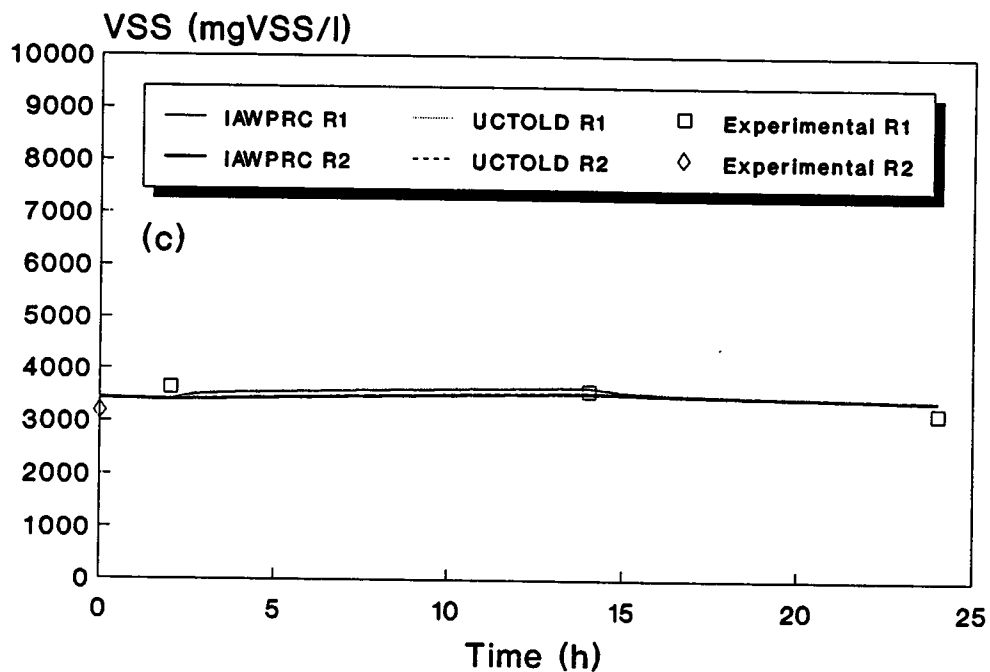
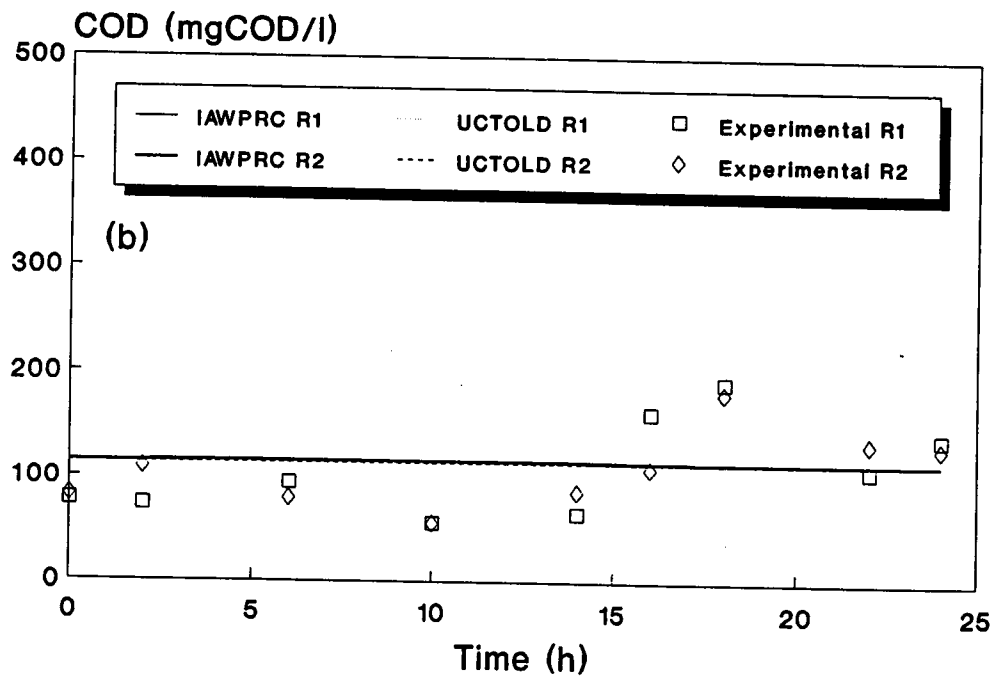


Fig 5.9: Predicted and experimental (a) OUR, (b) COD, (c) VSS, (d) TKN, and (e) nitrate time profiles for in-series anoxic (R1)/aerobic (R2) reactor system under cyclic flow and load (feed 12h on, 12h off); predictions using UCTOLD and IAWPRC, data from Wilson and Marais (1976). Continued overleaf.....

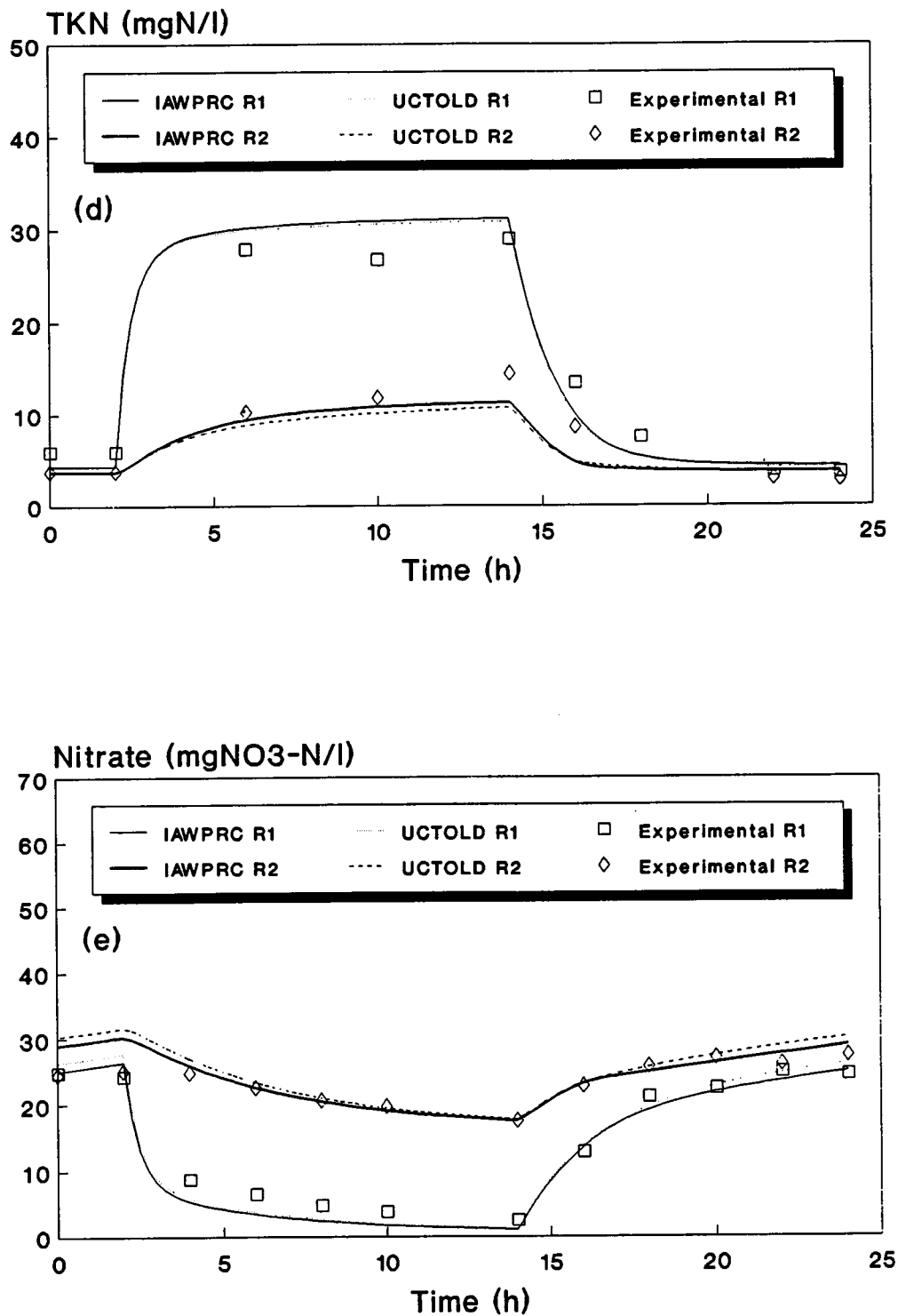
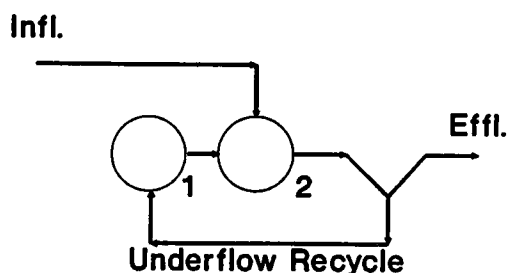


Fig 5.9: Predicted and experimental (a) OUR, (b) COD, (c) VSS, (d) TKN, and (e) nitrate time profiles for in-series anoxic (R1)/aerobic (R2) reactor system under cyclic flow and load (feed 12h on, 12h off); predictions using UCTOLD and IAWPRC, data from Wilson and Marais (1976).

CONTACT-STABILIZATION SYSTEM

A laboratory-scale investigation of the contact-stabilization system under constant and cyclic flow and load was conducted by Alexander, Ekama and Marais (1980) using unsettled wastewater from Cape Flats, Cape Town, South Africa.



1=Stabilization Reactor
2=Contact Reactor

Constant flow and load

Contact-stabilization systems at 6 and 10 days sludge age were operated. Data input for the 6 day sludge age system was as follows:

**** WASTEWATER CHARACTERISTICS ****		
Sti(avg)	g COD m-3	512.000
Nti(avg)	g N m-3	56.000
Fbs	g COD g-1 COD	0.230
Fs,us	g COD g-1 COD	0.100
Fs,up	g COD g-1 COD	0.130
Fn,a	g N g-1 N	0.750
Fnob,p	g N g-1 N	0.500
Fn,ous	g N g-1 N	0.030
Fs,zbh	g Zbh COD g-1 COD	0.000
VSS/TSS	g VSS g-1 TSS	0.750
Inf Alk	mole m-3	10.000
RETURN TO MENU		

Fbs = readily biodegradable/biodegradable COD
 Fs,us = frac infl COD that is sol unbiodegrad
 Fs,up = frac infl COD that is part unbiodegrad
 Fn,a = frac infl TKN that is NH3/NH4
 Fnob,p = frac organic bio N that is part
 Fn,ous = frac infl TKN that is organic unbio sol
 Fs,zbh = frac infl COD that is heterotrophs

Hit <Arrows>/<SpaceBar>
to move selection

Hit <Return> to enter
new value for
selected constant

**** PROCESS CONFIGURATION **** Gp 1. Number of Reactors = 2

Gp 2. Reactor Vols, l Gp 3. Feed Fraction Gp 4. Aeration/DO

No. 1:	12.30	0.00	3.0
No. 2:	2.00	1.00	3.0

Gp 5. Recycles: Use to include/remove mixed liquor recycles.

RAS recycle to Reactor No.1

Do you wish to change any parameters?

Y/N....

***** PLANT OPERATING PARAMETERS *****

1	SRT {total}	d	=	6.0
2	Process Temperature	degC	=	20.0

Flow rates:

3	Influent flow	l d-1	=	36.0
4	RAS recycle flow	l d-1	=	72.0

Do you wish to change any parameters?

Y/N....

Data input for the 10 day sludge age system was as follows:

**** WASTEWATER CHARACTERISTICS ****		
Sti(avg)	g COD m-3	496.000
Nti(avg)	g N m-3	47.000
Fbs	g COD g-1 COD	0.230
Fs,us	g COD g-1 COD	0.100
Fs,up	g COD g-1 COD	0.130
Fn,a	g N g-1 N	0.750
Fnob,p	g N g-1 N	0.500
Fn,ous	g N g-1 N	0.030
Fs,zbh	g Zbh COD g-1 COD	0.000
VSS/TSS	g VSS g-1 TSS	0.750
Inf Alk	mole m-3	10.000

RETURN TO MENU

Fbs = readily biodegradable/biodegradable COD	Hit <Arrows>/<SpaceBar> to move selection
Fs,us = frac infl COD that is sol unbiodegrad	
Fs,up = frac infl COD that is part unbiodegrad	
Fn,a = frac infl TKN that is NH3/NH4	
Fnob,p = frac organic bio N that is part	
Fn,ous = frac infl TKN that is organic unbio sol	Hit <Return> to enter new value for selected constant
Fs,zbh = frac infl COD that is heterotrophs	

**** PROCESS CONFIGURATION ****			
		Gp 1. Number of Reactors = 2	
Gp 2. Reactor Vols, l		Gp 3. Feed Fraction	Gp 4. Aeration/DO
No. 1:	12.30	0.00	3.0
No. 2:	2.00	1.00	3.0
Gp 5. Recycles: Use to include/remove mixed liquor recycles.			
RAS recycle to Reactor No.1			

Do you wish to change any parameters?
Y/N....

***** PLANT OPERATING PARAMETERS *****			
1	SRT (total)	d	= 10.0
2	Process Temperature	degC	= 20.0
Flow rates:			
3	Influent flow	l d-1	= 24.0
4	RAS recycle flow	l d-1	= 48.0

Do you wish to change any parameters?
Y/N....

With regard to the kinetic and stoichiometric constants, $\hat{\mu}_A = 0,66/d$ and $0,50/d$ for the 6 and 10 day sludge age systems respectively were necessary. The value of one other constant had to be changed from the default value – the half saturation constant for RBCOD (K_{SH}) had to be increased, from 5 to 110 mgCOD/l for UCTOLD and from 5 to 50 mgCOD/l for IAWPRC. The reasons for this change are detailed in the next section dealing with the contact-stabilization system under cyclic flow and load.

The predicted and simulated results are compared in Table 5.3.

Table 5.3: Comparison of measured and predicted results for contact stabilization system at 6 and 10 days sludge age receiving unsettled wastewater as influent.

Parameter	Sample Point	Sludge age : 6d			Sludge age : 10d		
		Measured	UCTOLD	IAWPRC	Measured	UCTOLD	IAWPRC
COD (mgCOD/ ℓ)	Contact	62	69	79	52	65	76
	Effluent	90	69	79	61	65	76
	Stabil.	54	53	62	42	51	59
TKN (mgN/ ℓ)	Contact	11	14	14	11	11	11
	Effluent	15	14	14	12	11	11
	Stabil.	1,7	3,5	3,5	4,8	3,1	3,1
NO ₃ (mgN/ ℓ)	Contact	27	28	28	22	24	24
	Effluent	27	28	28	23	24	24
	Stabil.	36	38	38	29	32	32
MLVSS (mgVSS/ ℓ)	Contact	1301	1540	1513	1478	1466	1439
	Stabil.	1772	2185	2148	2073	2080	2043
OUR (mgO/ ℓ /h)	Contact	61	61	59	42	43	41
	Stabil.	32	35	34	21	23	22

Cyclic flow and load

A contact-stabilization system under cyclic flow and load was operated at a sludge age of 5,8 days. The cyclic flow and load input data were as follows:

***** DIURNAL INPUT PATTERN *****				
Record No	Time (h)	Flow(l d-1)	COD (g m-3)	TKN (g m-3)
1	0.0	18.0	500.0	46.0
2	2.0	54.0	500.0	46.0
3	4.0	54.0	500.0	46.0
4	6.0	54.0	500.0	46.0
5	8.0	54.0	500.0	46.0
6	10.0	54.0	500.0	46.0
7	12.0	54.0	500.0	46.0
8	14.0	18.0	500.0	46.0
9	16.0	18.0	500.0	46.0
10	18.0	18.0	500.0	46.0
11	20.0	18.0	500.0	46.0
12	22.0	18.0	500.0	46.0

** Calculated Mean Values :	
Flowrate =	36.0
COD =	500.0
TKN =	46.0

Change any values? Y/N....

Wastewater characteristics and configuration and operational data input were as follows:

**** WASTEWATER CHARACTERISTICS ****	
Sti(avg) g COD m-3	500.000
Nti(avg) g N m-3	46.000
Fbs g COD g-1 COD	0.230
Fs,us g COD g-1 COD	0.100
Fs,up g COD g-1 COD	0.130
Fn,a g N g-1 N	0.750
Fnob,p g N g-1 N	0.500
Fn,ous g N g-1 N	0.030
Fs,zbh g Zbh COD g-1 COD	0.000
VSS/TSS g VSS g-1 TSS	0.750
Inf Alk mole m-3	10.000

RETURN TO MENU

Fbs = readily biodegradable/biodegradable COD Fs,us = frac infl COD that is sol unbiodegrad Fs,up = frac infl COD that is part unbiodegrad Fn,a = frac infl TKN that is NH3/NH4 Fnob,p= frac organic bio N that is part Fn,ous= frac infl TKN that is organic unbio sol Fs,zbh= frac infl COD that is heterotrophs	Hit <Arrows>/<SpaceBar> to move selection Hit <Return> to enter new value for selected constant
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**** PROCESS CONFIGURATION ****

Gp 1. Number of Reactors = 2

Gp 2. Reactor Vols, l

Gp 3. Feed Fraction

Gp 4. Aeration/DO

No. 1: 12.30

0.00

3.0

No. 2: 2.00

1.00

3.0

Gp 5. Recycles: Use to include/remove mixed liquor recycles.

RAS recycle to Reactor No.1

Do you wish to change any parameters?

Y/N....

***** PLANT OPERATING PARAMETERS *****

1	SRT (total)	d	=	5.8
2	Process Temperature	degC	=	20.0

Flow rates:

3	Influent flow	l d-1	=	36.0
4	RAS recycle flow	l d-1	=	72.0

Do you wish to change any parameters?

Y/N....

Sludge wastage to maintain the sludge age took place continuously over a 24 hour period from the contact reactor, as follows:

***** SLUDGE WASTAGE PATTERN *****

Record No	Time (h)	Wastage On	Waste Flow (l d-1)
1	0.0	Yes	3.42
2	2.0	Yes	3.42
3	4.0	Yes	3.42
4	6.0	Yes	3.42
5	8.0	Yes	3.42
6	10.0	Yes	3.42
7	12.0	Yes	3.42
8	14.0	Yes	3.42
9	16.0	Yes	3.42
10	18.0	Yes	3.42
11	20.0	Yes	3.42
12	22.0	Yes	3.42

** Mean wastage rate =

3.42 l/d

Switch any on/off? Y/N....

In the simulation, $\hat{\mu}_A = 0,55/\text{d}$. The value of one other constant had to be changed from the default value – the half saturation constant for RBCOD (K_{SH}) had to be increased, from 5 to 110 for UCTOLD and from 5 to 50 mgCOD/ℓ IAWPRC. This increase was necessary in order to give reasonable predictions of the soluble COD ($< 0,45\mu\text{m}$) in the contact reactor. No explanation for the increase is apparent at this stage – it is not certain whether the structure of the model is inadequate to accommodate the contact-stabilization system, or whether the increase in K_{SH} reflects a change in behaviour of the heterotrophic population when placed in the contact-stabilization environment in which the COD loading rate in the contact reactor is high. Accepting the increase in K_{SH} , predicted and measured results are compared in Fig 5.10(a,b,c,d and e); reasonable agreement is obtained.

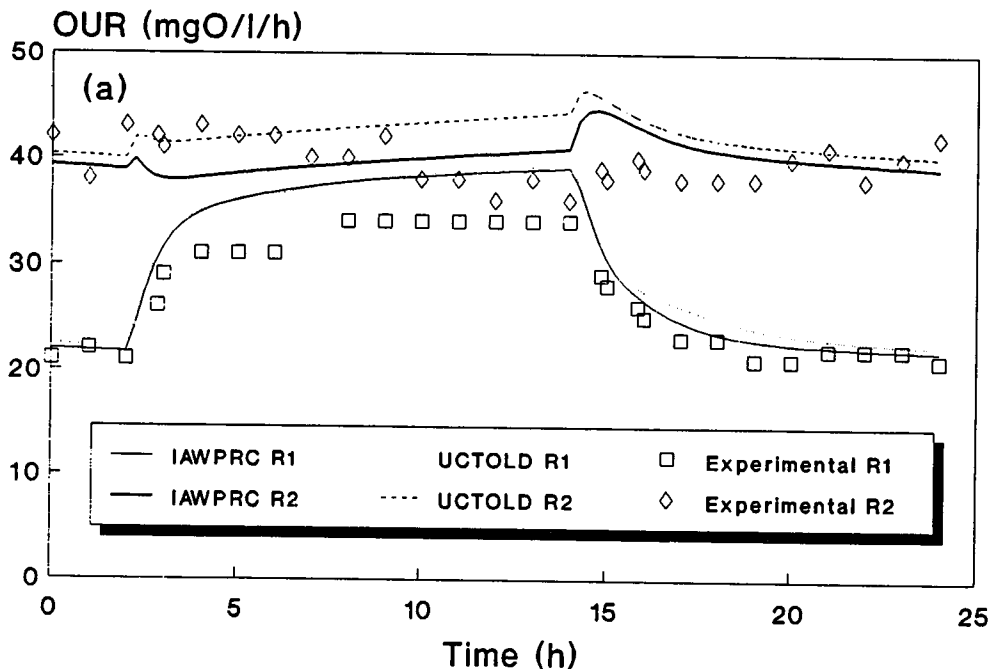


Fig 5.10: Predicted and experimental (a) OUR, (b) COD, (c) VSS, (d) TKN, and (e) nitrate time profiles for in-series aerobic contact (R2)-stabilization (R1) system under cyclic flow and load (feed 12h at $54\ell/\text{d}$, 12h at $18\ell/\text{d}$); predictions using UCTOLD and IAWPRC, data from Alexander *et al.* (1980). Continued overleaf.....

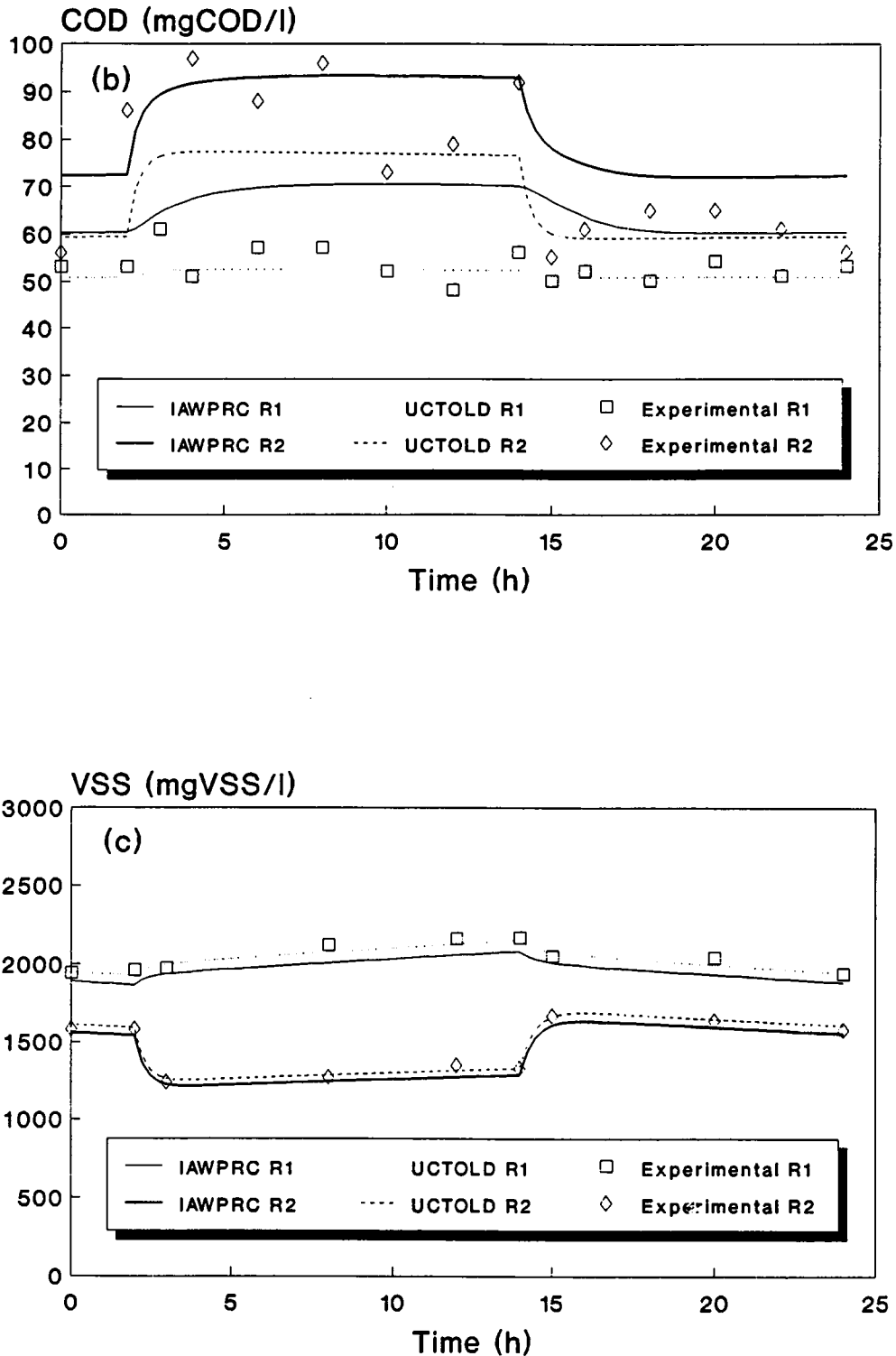


Fig 5.10: Predicted and experimental (a) OUR, (b) COD, (c) VSS, (d) TKN, and (e) nitrate time profiles for in-series aerobic contact (R2)-stabilization (R1) system under cyclic flow and load (feed 12h at 54l/d, 12h at 18l/d); predictions using UCTOLD and IAWPRC, data from Alexander *et al.* (1980). Continued overleaf.....

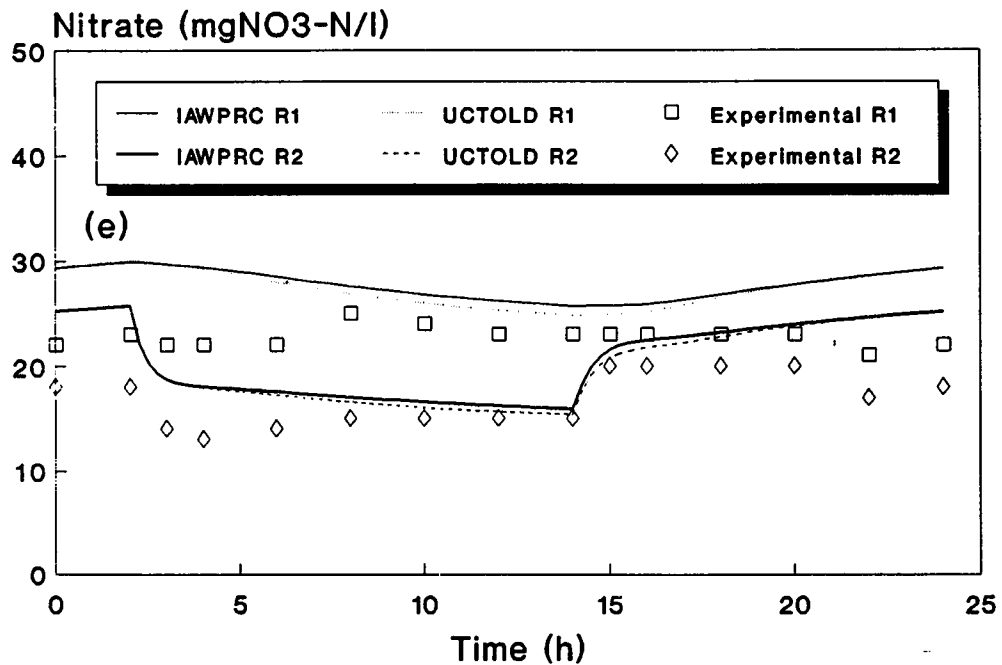
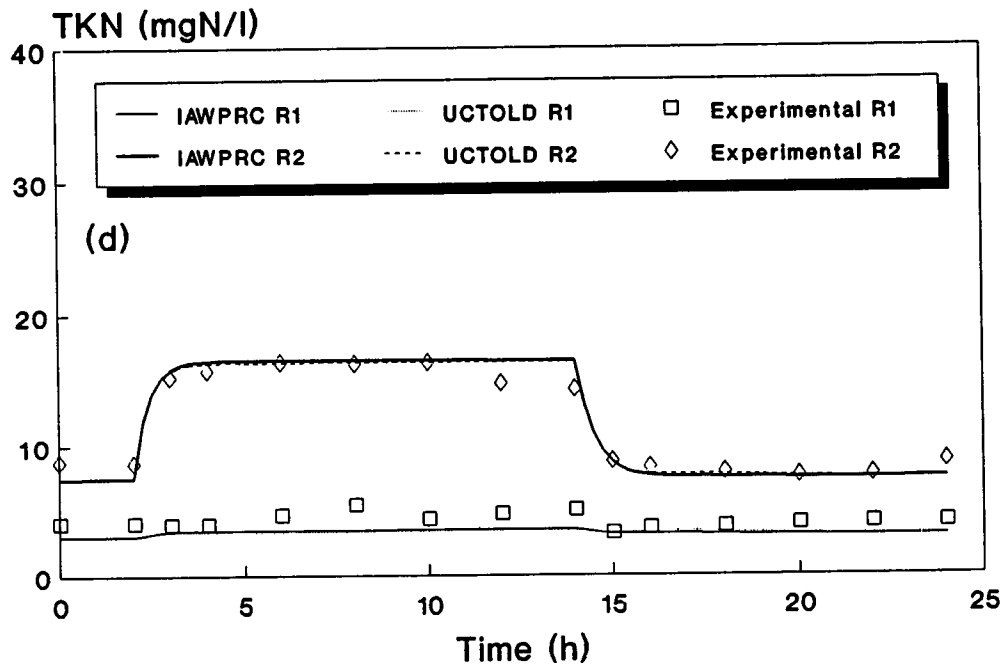


Fig 5.10: Predicted and experimental (a) OUR, (b) COD, (c) VSS, (d) TKN, and (e) nitrate time profiles for in-series aerobic contact (R2)-stabilization (R1) system under cyclic flow and load (feed 12h at 54l/d, 12h at 18l/d); predictions using UCTOLD and IAWPRC, data from Alexander *et al.* (1980).

CONCLUSIONS

From the simulations in this Chapter using the two models:

- Both models provide reasonable descriptions of the response of a wide range of systems and accordingly both can be accepted as adequate.
- Without further information it is not possible to pass judgement as to which model is to be preferred.
- For simulation of all systems, except for contact-stabilization systems and batch tests with nitrification inhibition, the default values for the kinetic and stoichiometric constants can be accepted with the exception of $\hat{\mu}_H$ and $\hat{\mu}_A$; $\hat{\mu}_H$ and $\hat{\mu}_A$ have been found to change with certain reactor configurations (e.g. selectors, plugflow) and influent load patterns (e.g. on-off feed pattern), $\hat{\mu}_A$ has been found to be dependent on characteristics of the wastewater feed also. (Procedures to determine these constants are set out in Chapter 6).
- For simulations of the contact-stabilization system, and batch tests in which nitrification has been inhibited by addition of thiourea, adjustment of a single constant is required in each case, the half saturation constant for RBCOD for the contact-stabilization system, and the half saturation constant for SBCOD hydrolysis for the batch test in which nitrification is inhibited.